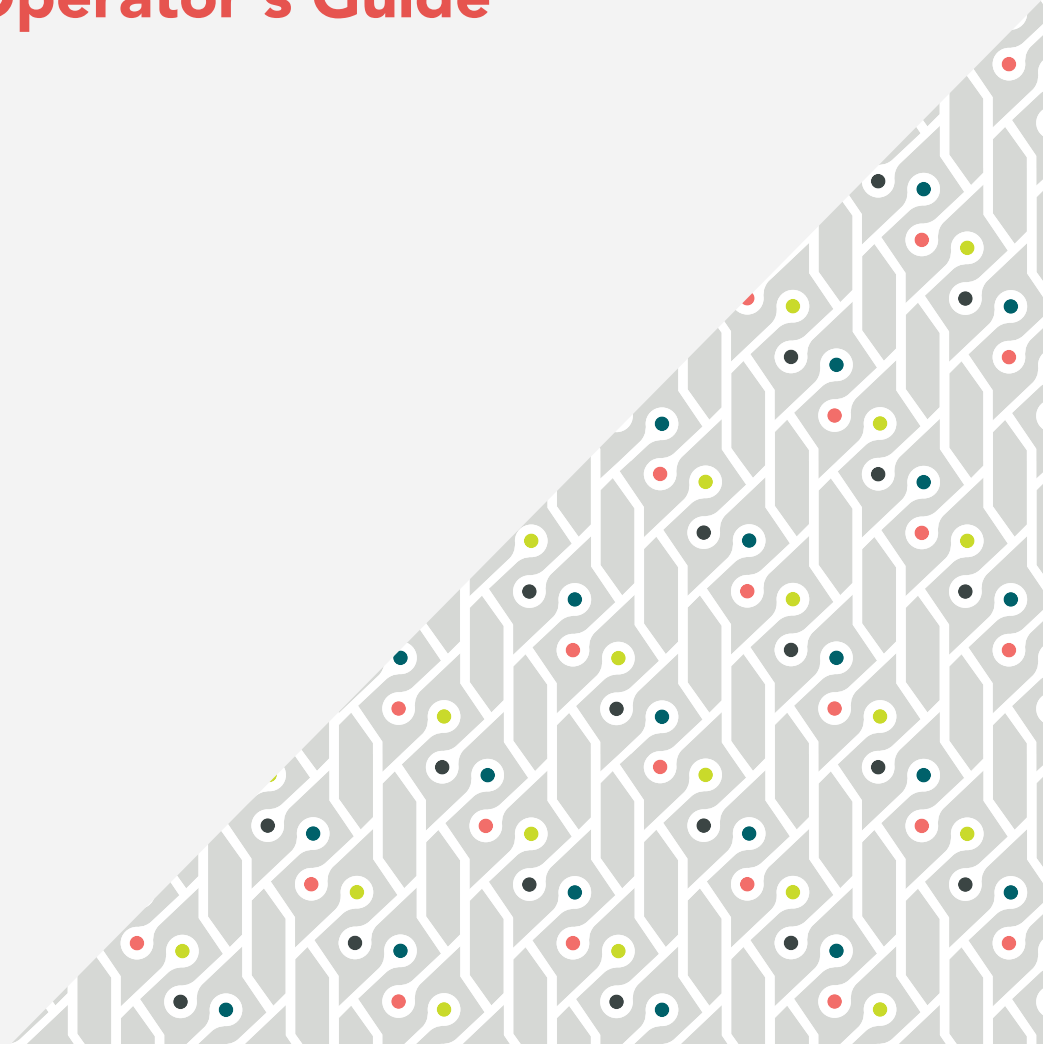




Kinarm End-Point Lab (PN 14332)

Operator's Guide





When installed in accordance with this manual and verified by a BKIN technician, the equipment described in this guide is in conformity with the relevant Essential Health and Safety Requirements of the following Directives:

- 2004/108/EC (relating to electromagnetic compatibility)
- 2006/95/EC (relating to electrical equipment designed for use with certain voltage limits)
- 2006/42/EC (on machinery)

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1 Revision History

Revision	Description	Date	Approved By
1	Initial release (based on Kinarm Operator Guide rev 9)	26-Mar-2021	IEB
2	Updates for Dexterit-E 3.10	15-Mar-2023	IEB
3	Updates for Dexterit-E 3.11.	31-Jul-2025	IEB
4	Update for new revision of Kinarm EP Lab 14332 that contains 15050 Power Console and AV Controller.	28-Oct-2025	IEB

2 Read This First

2.1 Product Disclaimer

Kinarm Standard Tests are currently in research and development and are currently only available to researchers with an IRB/REB approved protocol.

****Intended Use**

Kinarm Standard Tests are intended as research tools to contribute to the understanding of brain function and dysfunction. Kinarm Standard Tests are intended to be used on a Kinarm Exoskeleton Lab and/or a Kinarm End-Point Lab. Kinarm Standard Tests do not directly offer a medical diagnosis of any type, nor are Kinarm Standard Tests to be used as an assessment tool to assist with diagnosis. A diagnosis of any brain injury or disease can be made only by a qualified physician or psychologist.

****Contraindications**

Kinarm Standard Tests are not indicated for research subjects who do not have adequate cognitive functioning to understand the task instructions, who do not have visual and/or auditory acuity to permit adequate perception of instructions or task stimuli, and/or who are of an age falling outside of 18-85.

ATTENTION: Any components removed from a Kinarm Lab will not be supported for any other use.

2.2 Conventions

The user information that accompanies the product uses typographical conventions to assist you in finding and understanding information.

- All procedures are numbered and all sub-procedures are lettered. You must complete the steps in the sequence in which they are presented to ensure success.
- Bulleted lists indicate general information related to a function or procedure. They do not imply a sequential procedure.
- When a task has several approaches or option combinations these are indicated as choices. If applicable, the recommended option is listed first. Each choice completes the associated step.
- Control names and menu items or titles are spelled as they are on the system and they appear in highlighted text.
- Click or select means to move the pointer to an object or menu item and press the left mouse button.
- Right-click means to point at an item and then press the right mouse button.

- When an instruction involves menu-based actions, each menu item appears as **Menu Name**, child menus and options appear in order after the primary menu name. The series of menus and options appear as **File > Subject > Export Record**.

NOTE: Notes bring your attention to information that will help you better understand the product (i.e. for information purposes only).

ATTENTION: Attention notifications highlight information that will help you avoid use errors that are not related to safety.

CAUTION: Cautions highlight hazardous situations that, if not avoided, could result in minor or moderate injury.

WARNING: Warnings highlight hazardous situations that, if not avoided, could result in serious injury or death.

2.3 Audience

This information identifies the audience for the guide. These audience definitions are general in nature.

Prescriber

This individual is responsible for running Kinarm Standard Test assessments and interpreting the standard test reports.

Researcher

This individual is responsible for creating, running, and analyzing Custom Task Programs. This individual may also be responsible for interpreting the results of standard tests for research purposes.

Operator

This individual runs the Kinarm Standard Tests and the Custom Tasks. They are responsible for cleaning the Kinarm Lab after use.

Servicer

This individual is responsible for routine maintenance and, under the direction of Kinarm, repair/rework the Kinarm Lab as required.

2.4 Related Documentation

The different versions of Kinarm Labs that have been released over the years are identified with different part numbers (PN). Each version has its own unique documentation associated with it, so to ensure that you are using the correct documentation, please refer to [2.5 Identifying Kinarm Lab Versions](#).

Kinarm End-Point Lab (PN 14332) Hardware Guide

This guide describes the hardware components of the Kinarm End-Point Lab™ (PN 14332) made by BKIN Technologies Ltd. (dba Kinarm). The information in this guide includes the basic assembly and settings of the lab required for installation and configuration and lists the technical specifications of the Lab hardware.

Dexterit-E User's Guide

This guide describes the capabilities of the Dexterit-E™ software. The Dexterit-E software is used in conjunction with a Kinarm lab. Some basic information is shared in the lab's operator's guide to simplify operation. It is important to be familiar with the contents of this guide to understand the full range of features and functionality available in Dexterit-E.

This guide includes the user interface, upgrading the software, selecting and backing up databases, managing subject data, and running tasks. This guide describes the Kinarm Standard Tests and discusses creating Custom Tasks, task protocols, and analysis.

The guide includes information on calibrating and verifying the associated lab.

Other Available Guides

- **Adjustable Height Configuration Addendum for Hardware and Operator's Guides** - Provides a description of the hardware for the adjustable height configuration of the Kinarm End-Point Lab™.
- **Creating Task Programs** - Introduces the concept of Task Programs, the basic code structure of Task Programs, and demonstrates these with numerous examples.
- **Dexterit-E Explorer Users' Guide** - Provides instructions on using the Kinarm Dexterit-E Explorer™ program for viewing data collected in Dexterit-E.
- **Eyelink 1000+ User Manual** - Provides more detailed instructions on using the Eyelink 1000+ software, which is a core component of the Kinarm Gaze-Tracker.
- **Kinarm Analysis Scripts** - Provides a brief description of the various functions that can be used to load, manage, and analyze Kinarm data in MATLAB™.
- **Kinarm Standard Tests Summary** - Describes the Kinarm Standard Tests™ (KST) available as part of the Dexterit-E™ software usable on the Kinarm Lab.

- **Task Development Computer Setup Guide** - An excerpt from the Dexterit-E Addendum - Creating Task Programs that provides all necessary information for setting up the development environment to create Custom Task Programs for Dexterit-E.

2.5 Identifying Kinarm Lab Versions

The different versions of the Kinarm Exoskeleton Lab™ and Kinarm End-Point Lab™ that have been released over the years are identified with different part numbers (PN). Although overall functionality is the same for different versions, changes in hardware design have required some changes in appearance and operation. This section provides information that can be used to identify which revision of a Kinarm Lab you have.

Newer Kinarm Labs have labels indicating both a part number (PN) and a serial number (SN). If present, the PN label can be used to identify your Kinarm version. If not, then a SN Label may be of assistance - the first two digits of a SN Label indicate the year that the Kinarm was manufactured.

PN 10288. The first version of the Kinarm End-point Lab was released in 2009, and was in production until 2020. This version did not have a PN label for the Kinarm Lab, but each Kinarm Robot had its own PN label: PN 10212 for the right robot and PN 10213 for the left robot. If your Kinarm End-Point robots have individual Kinarm robot PN labels 10212 and 10213 and have a SN label starting with 20 or less, then the PN for your Kinarm Lab is PN 10288.

PN 14332. The second version of the Kinarm End-point Lab was released in 2021. It has a PN label on the back of the right Kinarm Robot indicating PN 14332.

ATTENTION: In late 2025, a new version of the Kinarm End-Point Lab PN 14332 was released, containing Power Console PN 15050 and an AV Controller. Rev 2 of the hardware guide applies to this new revision. Older Kinarm End-Point Labs PN 14332 should continue to use Rev 1 of the hardware guide as the main reference, but can reference Rev 2 for information on features not covered by Rev 1.

2.6 Warnings and Cautions

Custom Task Creation

CAUTION: Ensure the emergency stop button is readily accessible and/or keep a firm grip on the robot(s) when testing newly edited Custom Task Programs. Applying new loads can result in unexpected behaviour such as excessive forces and/or unstable loads. Care must be taken to reduce the risk of injury to the subject or operator, and/or damage to the Kinarm Lab.

Electrical and Heat Safety

WARNING: To avoid the risk of electric shock, the Kinarm Lab must be connected to a supply mains with protective earth.

WARNING: To reduce the risk of electric shock, do not connect anything that is not part of the Kinarm Lab to the Power Console.

WARNING: To avoid the risk of electric shock, do not remove enclosure panels.

WARNING: To avoid the risk of electric shock, keep water and other liquids away from the Kinarm.

Hygiene

CAUTION: Clean all components of the system that come into contact with subjects between subjects to reduce the risk of pathogen transfer.

General Lab Setup and Use

CAUTION: Do not allow bystanders within the robot workspace. Ensuring that bystanders remain outside of the robot workspace at all times reduces the risk of injury.

CAUTION: Do not leave a harnessed subject unsupervised. Subjects should be under the constant supervision of a trained and qualified operator at all times.

CAUTION: Do not alter or remove any hardware from the lab. Hardware that requires the use of a tool to remove or adjust should not be detached or repositioned unless under the direction of Kinarm Support. Leaving the hardware as originally installed reduces the risk of subject injury. This includes ensuring the safety stops are not removed.

CAUTION: Do not install unsupported software/OS on the Kinarm Lab computers. Avoid running any program simultaneously with Dexter-E (such as virus scans, web browsers, copying files, email, watching movies, etc.) to reduce the risk of degrading system performance.

CAUTION: Do not overload the Power Rack drawer. Keeping the total mass of items in the drawer to a maximum of 5 kg reduces the risk of the Power Rack tipping.

2.7 Updates and Upgrades

Kinarm is committed to innovation and continued improvement. When upgrades that consist of hardware or software improvements are released, updated user documentation will accompany those upgrades.

Software downloads for your Kinarm Lab can be found at:

- <https://kinarm.com/learning-support/software-downloads/>

Documentation for your Kinarm Lab can be found at:

- <https://kinarm.com/learning-support/user-guides-documentation/>

2.8 Customer Service

If you encounter problems with your Kinarm Lab or have questions about the user documentation, contact Kinarm Support at:

- support@kinarm.com

If applicable, generate and include a log file from Dexterit-E (**Help > Generate Log**; see the Dexterit-E User Guide for more details).

3 In This Guide

Welcome to the Kinarm End-Point Lab (PN 14332) Operator Guide. This guide is intended to describe the steps necessary to run a subject through a task from start to finish in a Kinarm End-Point Lab™ with part number 14332 using Dexterit-E™ and optionally Kinarm Standard Tests™. This guide assumes that the Kinarm Lab has been correctly installed and verified. For more details about the hardware and its installation, refer to the appropriate Kinarm Lab Hardware Guide. Further details about Dexterit-E and its setup can be found in the Dexterit-E User Guide.

The Kinarm End-Point Lab (PN 14332) Operator Guide provides step-by-step instructions for performing a subject exam using the Kinarm Lab. Instructions on using the optional Integrated Kinarm Gaze-Tracker™ module are also included. Each chapter deals with a different aspect of using a Kinarm Lab, including what to do before, during and after a subject exam, as well as annual maintenance checks.

ATTENTION: The different versions of Kinarm Labs that have been released over the years are identified with different part numbers (PN). Each version has its own unique documentation associated with it, so to ensure that you are using the correct Operator Guide, please refer to [2.5 Identifying Kinarm Lab Versions](#)

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4 Using a Kinarm Effectively

This chapter reviews a few of the concepts that are important to use a Kinarm effectively.

4.1 Factors for Subject Screening

The Kinarm Lab is designed to accommodate a range of body sizes, weights and impairments, and a minimum level of cognitive capabilities. The morphometric range includes the 5th-95th percentile of the general population from 5 years of age. The range is based on North American anthropometric data.

ATTENTION: Subjects outside of the designed range may not fit comfortably in the Kinarm Lab.

ATTENTION: The Kinarm End-Point Lab with a Virtual Reality Display will fit small adults and teenagers comfortably. Younger, smaller subjects may need to kneel.

Specific limitations include:

- **Weight limits:** the maximum weight is:
 - 280 lbs (127 kg) for the Standard Chair
 - 330 lbs (150 kg) for the Deluxe Chair
- **Minimum function for Kinarm Standard Tests:** research subjects must be able to read at a fifth-grade reading level, have adequate cognitive functioning to understand the task instructions, and have visual and/or auditory acuity to permit adequate perception of instructions or task stimuli.

4.2 Subject Preparation for Robust Gaze-Tracking

The Kinarm Gaze-Tracker determines a subject's gaze by analyzing video data of the subject's eye and a target sticker that is placed on the subject's face. In order to function properly, the Gaze-Tracker needs a clear view of both the subject's eye and the target, without any distractions. Because the Gaze-Tracker operates in the infrared spectrum, objects that are bright or reflective in the infrared spectrum can create problems. The initial preparation steps described below are required to ensure that once the subject is wheeled up to the Virtual Reality stand, the Gaze-Tracker will have the necessary clear view of the subject's target sticker and eye.

For the Gaze-Tracker camera to have a clear view of the target sticker, the target sticker needs to be placed in an appropriate location. Ideally, the sticker should be placed 0-3 cm off the mid-line just above the eyebrows. Alternatively, the sticker may be placed on the subject's cheek, just below the eye being tracked. However, care should be taken with this approach.

Due to the nature of the cheek muscles, changes in a subject's facial expression during a task could move the target and cause gaze tracking problems.

ATTENTION: If the target sticker is too far off mid-line, then it might be at too oblique an angle to be detected properly when the subject rotates their head.

ATTENTION: If the target sticker is too high on the subject's forehead, it might get blocked from the camera's view by the Virtual Reality display, and/or depending on the shape of the subject's head the sticker might be at too oblique an angle to be detected properly.

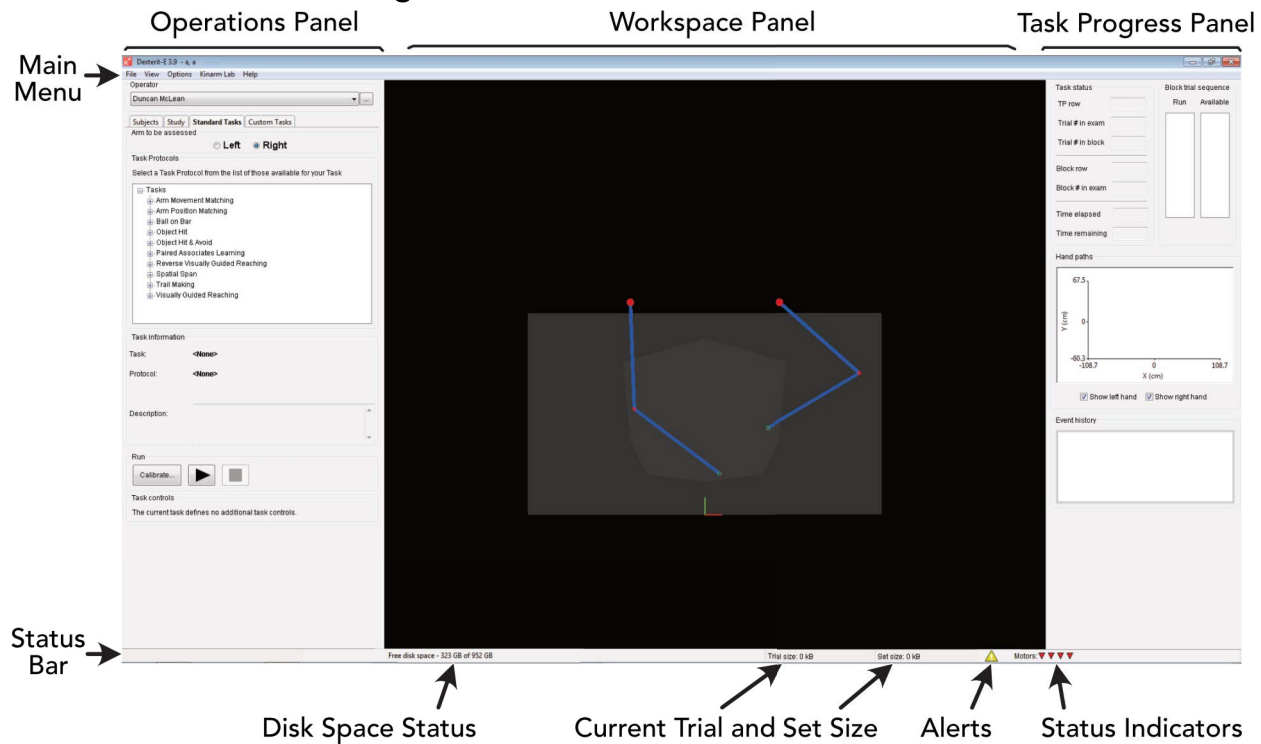
Common distractions and obstructions that can affect the Gaze-Tracker include:

- Eye makeup (e.g. mascara, eye-liner and/or eye shadow)
 - It is helpful to keep makeup remover wipes available for subjects wearing eye makeup.
- Other face-coverings (e.g. masks)
- Stray hairs, particularly for subjects with long hair
- Long eyelashes
- Lower and/or upper eyelids
 - If lower eyelids are causing problems, having the subject tilt their head will usually rectify the problem.
 - If upper eyelids are causing problems, it may be necessary to gently 'pull' up the subject's upper eyelid so that it is pulled open wider than normal. Surgical tape can be used for this purpose. If tape is used in this manner, ensure that the subject can still blink without discomfort.
- Eyeglasses (e.g. with reflective rims, small frames, or coated lenses that block infrared light)
 - If eyeglasses are a problem, consider removing them if the subject can see the targets well enough without them. Otherwise, it might be necessary to obtain a set of wide-framed, anti-reflective eyeglasses with interchangeable lenses for the subject to use (e.g. from [Solo Bambini](#)).

4.3 Dexterit-E User Interface

Operators use Dexterit-E to interact with the robots, as shown in the figure below.

Figure 4.1: Dexterit-E User Interface



Operators can modify the user interface via several methods:

- The view in the workspace panel can be:
 - tilted by dragging the mouse on the window;
 - zoomed using the mouse wheel; or
 - shifted by holding down the mouse wheel and dragging.
- The two standardized views are overhead (straight down) and perspective (similar to what the subject would see).
- A simplified view of the *Task Progress Panel* is available. See [Table: 4.1](#) for a comparison of the standard and simplified views.

Table 4.1: Task Progress Panel Options (Sheet 1 of 2)

Display Item	Simplified	Standard
Count of Trials	•	•
Elapsed Time	•	•
Last Event	•	•
All Events		•
Hand Paths		•
Trial Protocols In Current Block		•

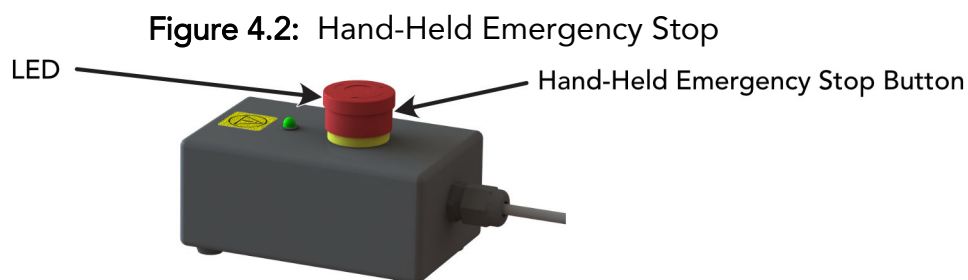
Table 4.1: Task Progress Panel Options (Sheet 2 of 2)

Display Item	Simplified	Standard
Trial Count		•
Trial Protocol		•
Block ID and block count		•
Elapsed and Remaining Time		•

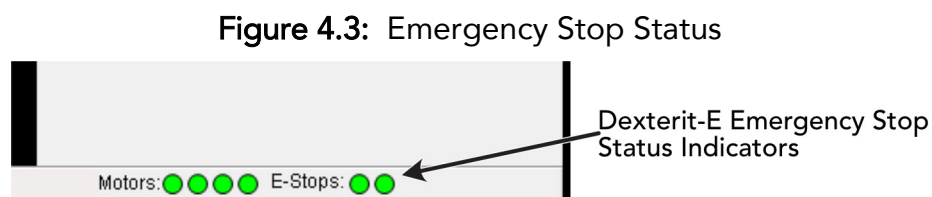
For further details on the Dexterit-E User Interface, refer to the Dexterit-E User Guide.

4.4 Emergency Stops

The Kinarm Lab comes with an emergency stop which must be enabled for the motors to be enabled and produce torques. This hand-held emergency stop button is portable (see [Figure: 4.2](#)). This button serves to enable or disable all motors in the Kinarm Lab. The LED in the hand-held emergency stop button enclosure turns green when the button is enabled, and red when disabled. Push the button down to disable the motors. Pull the button up or rotate it until it pops back up to re-enable the motors.



The state of the emergency stop system is indicated in the status bar in Dexterit-E (see [Figure: 4.3](#)). If a status indicator is red, then the emergency stop is engaged.



4.5 Study Protocols

To ensure consistency and reduce errors when running studies, Dexterit-E uses Study Protocols. Study Protocols are made up of:

- a list of sessions in the study (e.g. Intake visit, 4 month follow-up).

- an ordered list of Task Program/Task Protocol pairs to run each session.

Sessions and Task Program/Task Protocols are visible in the *Study* tab when a particular study is active. To create or edit a Study Protocol, refer to the Dexterit-E User Guide.

When Kinarm Standard Test data are collected as part of a Study Session it is possible to use Dexterit-E Explorer to create a Session report. If multiple Study Sessions are run on a single subject then Dexterit-E Explorer can be used to create Longitudinal reports.

4.6 Custom Task Protocols

Custom Task protocols are sets of parameter values that are used in conjunction with a Task Program to define and control system behaviour. The parameters available are defined by the Task Program, and are mostly unique to each Task Program (i.e. different Task Programs will have different sets of parameters available).

For information on coding a Custom Task, see the Dexterit-E addendum Creating Task Programs.

4.6.1 Create or Edit a Custom Task Protocol

NOTE: Task Protocol parameters cannot be edited while a task is running or paused. A task must be newly loaded or reset for the Task Protocol to be editable.

Figure 4.4: Edit a Task Protocol in Dexterit-E

The screenshot shows the 'Edit task protocol (100s)' dialog box. It features several tabs: General, Target table, Load table, TP table, Task wide params, Block table, and Analog inputs. The 'General' tab is selected. Inside, there are sections for 'Task instructions' with 'Default instructions' and 'Customized instructions' text areas, and a 'Video' dropdown. Below is 'General protocol information' with 'Code' and 'Description' text boxes. Then, 'Hand feedback behaviour' shows 'Selected behaviour' as 'Both hands' with a note 'Feedback will be shown for both hands.' 'Gaze feedback behaviour' shows 'Selected behaviour' as 'Off' with a note 'Gaze_Feedback block not found in model.' 'Pause/unpause behaviour' shows 'Selected behaviour' as 'Pause at next trial' with a note 'Pause at the beginning of the next trial. Pressing the Run/Pause button again continues the next trial.' 'Advanced options' includes a note 'The repeat trial flag has been disabled in the current task program.', a checked 'Use random seed' checkbox, and a 'Data rate' dropdown set to '1000 Hz' with a note 'Samples per second saved'. At the bottom are buttons for 'Apply', 'Apply and Save', 'Apply and Save As...', and 'Cancel'.

1. Select the *Custom Tasks* tab.
2. Select the desired Task Program.
3. Select one of the following:
 - To create a new Task Protocol, click the **Create new protocol** button.
 - To edit an existing protocol, select the desired Task Protocol and either click the **Edit selected protocol** button or right-click on the Task Protocol and select **Edit** from the context menu.

The *Edit task protocol* dialog opens (Figure: 4.4).

4. Enter/modify the parameter values as desired.
 5. Save the Task Protocol.
- If this is a new Task Protocol, it will now appear in the Task Protocol list.

4.6.2 Task Protocol Parameters

Values for the various parameters must be entered into the *Edit Task Protocol* dialog ([Figure: 4.4](#)). There are eight tabs in this dialog:

- General
 - Defines the parameters and settings that apply generally to the task
- Target Table
 - Defines the targets to be used during the task
- Load Table
 - defines the loads to be used during the task

CAUTION: Ensure the emergency stop button is readily accessible and/or keep a firm grip on the robot(s) when testing newly edited Custom Task Programs. Applying new loads can result in unexpected behaviour such as excessive forces and/or unstable loads. Care must be taken to reduce the risk of injury to the subject or operator, and/or damage to the Kinarm Lab.

- Task Protocol Table
 - defines the behaviour of all trials to be used during the task
- Task-Wide Parameters
 - defines any parameters used across all trials in the task
- Block Table
 - defines the number and sequence of trials which occur during the task
- Analog Inputs
 - defines and identifies the analog channels recorded in the data files
- Input Files
 - defines additional file(s) to be used by the Task Program
 - the fields on this task will only be enabled if the Task requires them

For each of the tables, each row details a different instance of that element (e.g. each row of Target Table is for a different target). Each column refers to a parameter that needs to be specified for the element. Additionally, for each of the tables excluding the Block Table, different tasks will typically have different columns. See the Dexter-E User Guide for more details.

Once a Task Protocol has been specified, it is advisable to save it for later use.

ATTENTION: Verify the units used with any parameter.

5 Lab Preparation

It's important that you understand how to start each component of the Kinarm lab. In addition, some Kinarm robots and various peripherals may require daily or subject-specific calibrations. If you are not familiar with the various components of the lab, refer to the information in the hardware guide accompanying your system.

Subjects, particularly those with mobility issues, are best accommodated when the chair is initially located at a neutral starting point away from other equipment.

5.1 Power On the Kinarm Lab

The Kinarm lab is composed of a series of integrated components, which need to be powered on and initialized.

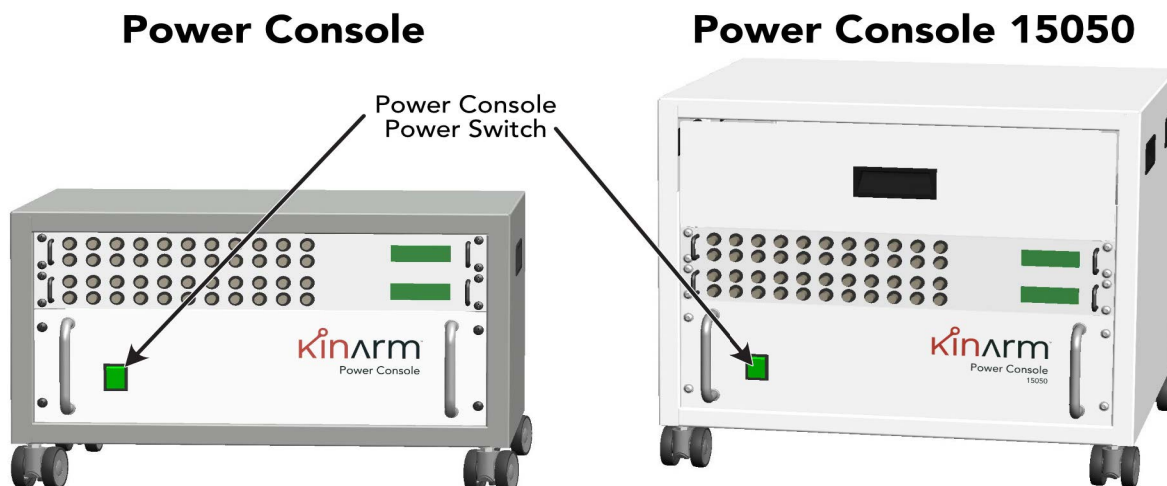
For newer Kinarm Labs with Power Console 15050 (as indicated on the front panel of the Power Console), the On/Off switch on the Power Console controls the On/Off state of a number of Kinarm components, including the Kinarm robots and the Robot Computer. For older Kinarm labs, the Power Console only controls the On/Off state of the Kinarm robots. See the applicable Kinarm Hardware Guide for details.

1. Start the following equipment:

- a. Power on the Power Console.

The power switch is on the front of the Power Console, as shown in the figure below.

Figure 5.1: Kinarm Lab Power Switch Location



The power switch lights up when the Power Console is powered on.

- b. Power on the Dexterit-E Computer.

NOTE: If you do not see a Windows screen on the Operator Display, ensure the Operator Display is on.

NOTE: For older Kinarm labs that have a KVM switch, tap the **Scroll Lock** key twice rapidly on the keyboard to switch the operator display between showing the Robot Computer vs. the Dexterit-E Computer.

- c. For Kinarm Labs with the older Power Console, power on the Robot Computer. For newer Kinarm Labs with Power Console 15050, the Robot Computer will power on when the Power Console is powered on.

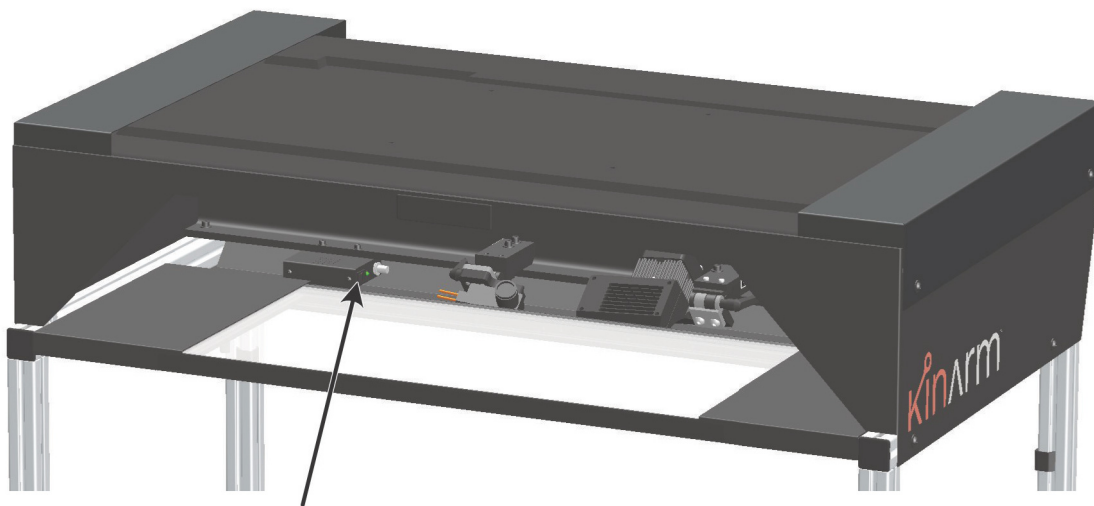
2. Log into Windows on the Dexterit-E Computer.

3. Look into the Virtual Reality Stand to confirm that the subject display is on. If the Kinarm screensaver is not visible, then power on the subject display manually.

4. Prepare the optional Gaze-Tracker, if the Gaze-Tracker is part of the lab configuration.

- a. Power on the Gaze Power Box.

Figure 5.2: Gaze-Tracker Power Switch Location



Gaze-Tracker Power Switch

- b. Power on the Gaze-Tracker computer.
- c. Start the EyeLink 1000 Host Application.
 - If given the option by the Gaze-Tracker computer during power-on, select the EyeLink operating system to boot. EyeLink 1000 Plus Host Application starts automatically.
 - If the Gaze-Tracker computer is already on, and the SR Research EyeLink File Manager is running, click the EyeLink icon on the left of the toolbar to start the EyeLink 1000 Plus Host Application.
- d. Remove the lens cover from the Kinarm Gaze-Tracker camera.

5. Disconnect the charging cable from the optional Kinarm Deluxe Chair if the chair is part of the lab configuration.
6. Start Dexterit-E.
7. Select the operator in Dexterit-E.
If the current operator is not present in the list, then first add them to the list.

5.2 Calibrate Robot Joints

Unlike older Kinarms, the Kinarm End-Point lab (PN 14332) does not require any robot joint calibration steps even if the robots or computers are powered down.

NOTE: The *Calibration* dialog indicates with large icons whether or not a particular calibration step is required, and what its status is. If a green check mark appears beside a calibration step, that calibration step has been completed and does need not be repeated.

5.3 Prepare the Lab for Subject Transfer

Before a subject arrives, operators can adjust the chair height and location to make the lab as accessible as possible. Optimally, the initial height and location minimize the effort required later for chair adjustments.

With the Standard Chair, the goal is to have the chair high enough so that it does not need to be raised after the subject sits down. It is simple to lower the Standard Chair if the subject's legs do not fit, but it is difficult to raise the seat without getting the subject to stand up. A good starting point is to set the chair's seat height so that the chair fits under the table top with about 2-3 inches (5-7 cm) to spare (see [Figure: 5.3](#)). Its wheels lock in a manner that is typical for office chairs.

With the Deluxe Chair, the goal is to set it at a height that will be safe and comfortable for the upcoming subject to sit in. The deluxe chair is motorized and so its height can be easily adjusted up or down after the subject is seated. In addition, it has a subject belt and harness, and a single pedal for locking the wheels (see [Figure: 5.4](#)).

Prior to subject arrival, the operator will have a choice of how to prearrange the harness and belt, depending on whether or not the operator thinks that the harness and belt will be used. Typically, the harness is used to provide trunk support for subjects with an impairment.

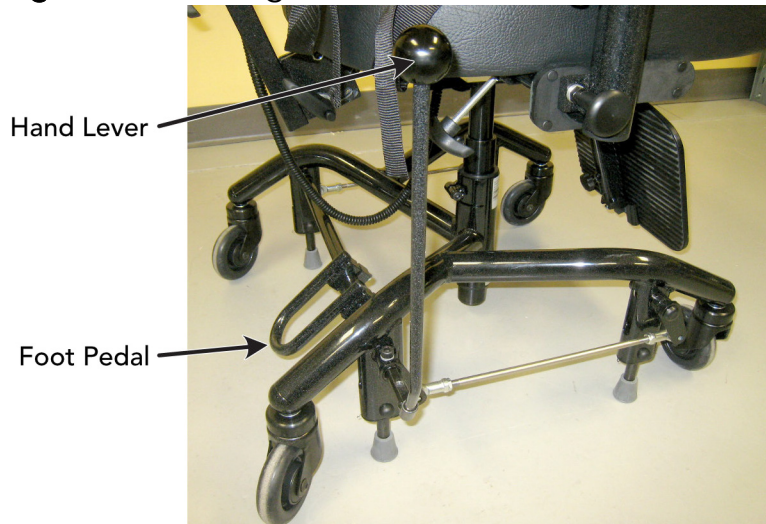
A full description of how to use all features and adjustments of the chairs can be found in the Kinarm End-Point Lab Hardware Guide.

CAUTION: To reduce the risk of injury during subject transfer and chair adjustments, remember to lock the chair wheels.

Figure 5.3: Set the Standard Chair's Height for the End-Point Lab



Figure 5.4: Locking the Wheels on the End-Point Lab Deluxe Chair



1. Adjust the seat height.

ATTENTION: If the up/down adjustment on the Deluxe Chair does not function, the chair may have entered stand-by mode. It can be taken out of stand-by mode by plugging the charger into the charging port for a few seconds and then unplugging it.

2. Move the chair a moderate distance away from the Virtual Reality Stand.
The goal is to have the chair far enough away from the Virtual Reality Stand that the subject can seat themselves without bumping into any equipment.
3. Lock the chair wheels.

4. If using a Deluxe Chair, ensure the armrests are locked in the upright position (i.e. the fold-away locking pin is locked).
5. If using a Deluxe Chair, fold up the footrest to prevent tripping or stepping on it.
6. If using a Deluxe Chair, prepare the harness and belt:
 - If the harness and belt are not going to be used:
 - Tuck the belt to the back of the seat, preferably under the backrest, so the subject does not sit on the belt
 - Unclip the harness at the chest clip and tuck the harness sides behind the backrest so the subject does not lean against the harness
 - If you expect to use the harness and belt:
 - Loosen the adjustments on both the belt and the harness, it is typically faster to shorten the straps if they are too long than lengthen them if they are too short
 - Unclip the belt and let the sides dangle off the sides of the chair for easy access
 - Unclip the harness at the chest strap and tuck each side of the harness behind the backrest so they are easy to access
7. If applicable, close window blinds/curtains.

ATTENTION: In rooms with lots of ambient light, it is recommended that window blinds/curtains be closed. This will ensure that the subject can see properly in the Virtual Reality Display.

6 Subject Preparation

After the subject has arrived and been screened, they must be properly set up in the Kinarm Lab.

6.1 Screen Subjects

When subjects arrive at the lab, it is important that the operator assess the subject to ensure that the lab can accommodate them and that they are prepared for the session. For the limits on the subject, see guidelines in [4.1 Factors for Subject Screening](#).

1. Review the subject's capabilities/impairments per the guidelines.
2. Review the subject's physical attributes per the lab limits.
3. Ensure the subject is able to understand the instructions for using the lab to accomplish the tasks assigned for the session.

6.2 Set Up a Dexterit-E Subject Record

A list of all subjects in the database is found in the Subject list of **Subjects** tab. See [Figure: 6.1](#). Each Windows login will typically have a unique subject database. By creating different Windows logins, restrictions to subject datasets may be managed for compliance with ethics protocols. It is important to ensure that the correct subject information is entered into Dexterit-E prior to collecting data.

At any point in time (i.e. during subject creation or during a subsequent edit) a subject can be assigned to a specific study. When a subject is assigned to a specific study, that study will be the default study for the subject for all subsequent visits.

Figure 6.1: Subjects Tab in Dexterit-E

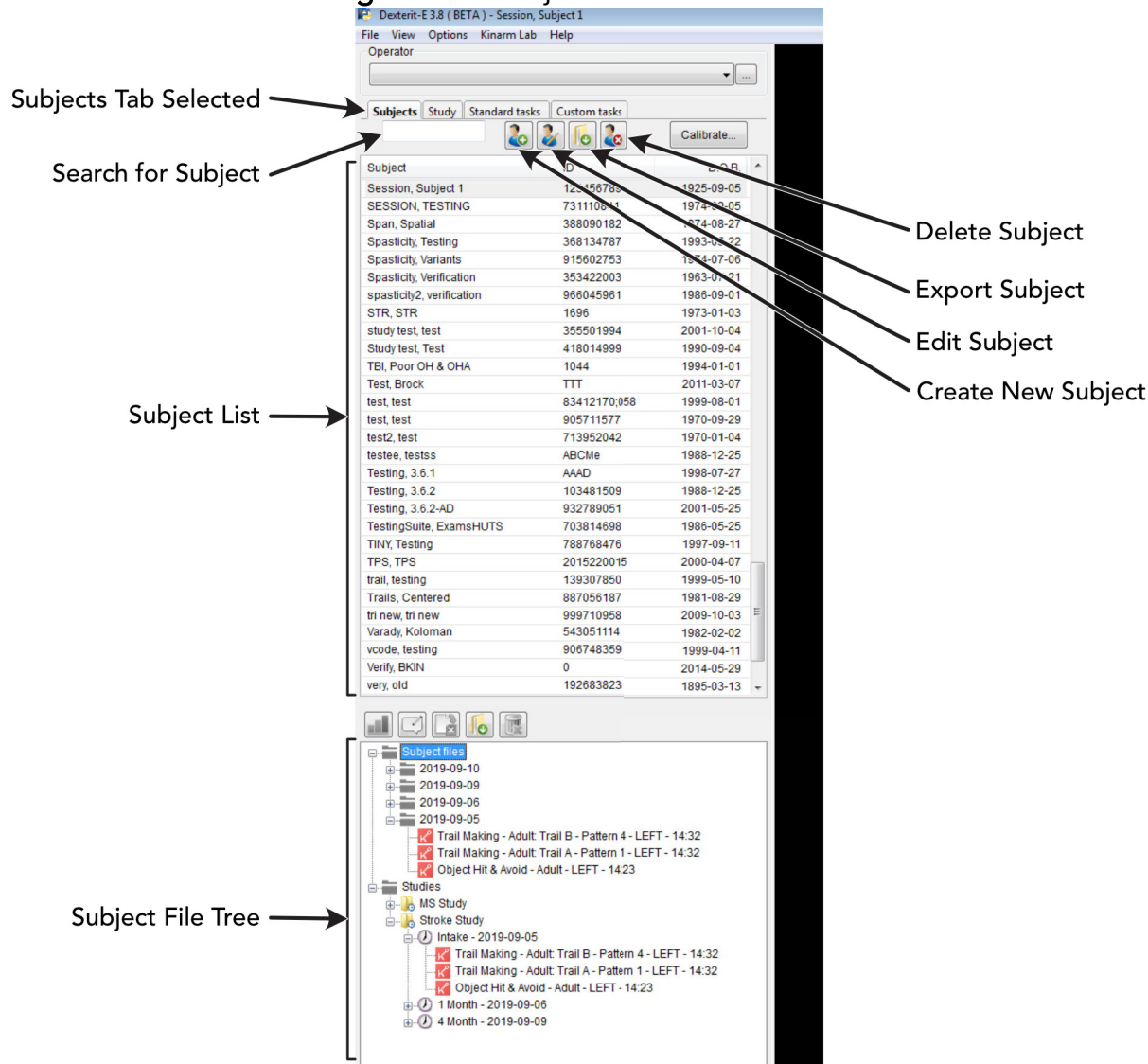


Figure 6.2: Create/Edit Subject Dialog in Dexterit-E

The dialog box is titled "Create new subject". It contains the following fields and controls:

- ID** (required): Text box containing "773967114".
- Last name** (required): Empty text box.
- First name** (required): Empty text box.
- Date of birth** (required): Calendar icon and text box containing "5/10/2018".
- Sex**: Radio buttons for M (selected), F, and Other.
- Height/Weight** (required): Text boxes for "0.0" cm and "0.0" kg.
- Handedness** (required): Radio buttons for Left and Right (selected).
- Notes**: Large text area for additional information.
- Study**: Dropdown menu.
- Pathology**: Dropdown menu.
- P.I.**: Dropdown menu.
- Create** and **Cancel** buttons at the bottom right.

If the subject is not already in the database, a new subject record must be created prior to running a task. If the subject is already in the database, the subject must be selected and verified prior to running a task.

1. Select the *Subjects* tab (see [Figure: 6.1](#)).
2. Click the **Create Subject** (or **Edit Subject**) button (see [Figure: 6.1](#)).
3. Enter (or edit) the subject information in the *Create new subject* dialog (see [Figure: 6.2](#)). All dialogue items marked in red are required; subject IDs must be unique.
4. Click **Create** (or **Save**).

6.3 Prepare the Subject for the Kinarm Gaze-Tracker

The Kinarm Gaze-Tracker determines a subject's gaze by analyzing video data of both the subject's eye and a target sticker that is placed on the subject's face. In order to function properly, the Gaze-Tracker needs a clear view of both without any distractions.

Ideally, the target sticker should be placed 0-3 cm off the mid-line just above the eyebrows. Alternatively, the sticker may be placed on the subject's cheek, just below the eye being tracked.

ATTENTION: If the target sticker is too far off mid-line or too high on the subject's forehead, then it might be at too oblique an angle to be detected properly when the subject moves their head.

Potential issues can be tested more thoroughly after the subject is set up by viewing the images of the subject's eye with Gaze-Tracker Computer's *Camera Setup* screen (see [Figure: 7.3](#)).

1. If the subject has eye makeup on, have them remove it.
2. Add the target sticker.
3. Clear away obstructions (e.g. hair, eyeglasses, eyelashes and/or eyelids).
Please see [4.2 Subject Preparation for Robust Gaze-Tracking](#) for further information on clearing obstructions.

6.4 Transfer the Subject to the Chair

If capable, the subject can seat themselves in the chair. If assistance is required, the operator and/or any other properly trained technician(s) should assist the subject with the transfer. The subject should be comfortable and centred (left/right) in the chair.

If using the Deluxe Chair, there are various features that can be adjusted and used for both subject comfort and improved usability:

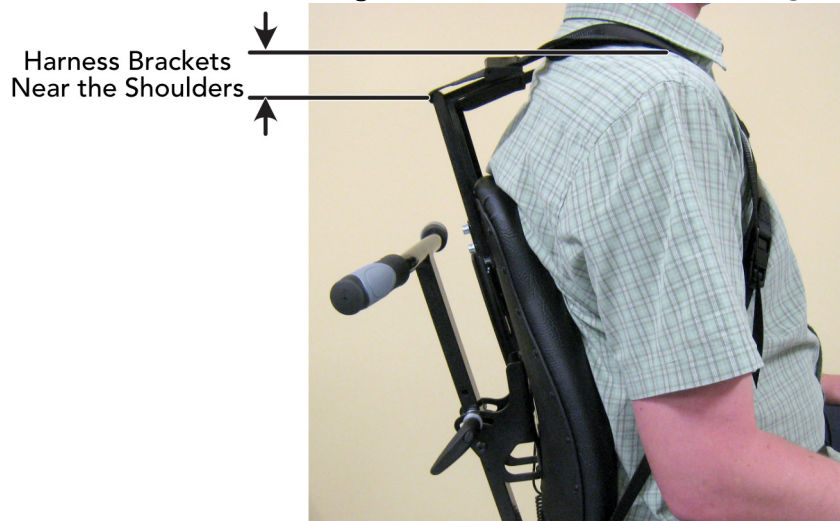
- Armrests (height, mediolateral position and fold-away).
CAUTION: Whenever the armrests are moved back to the upright position, the operator should ensure that the fold-away locking pin is locked back in place. If the armrests are not locked in place, the armrest may fold away if it is leaned on.
 - Backrest (height and angle).
 - Footrest (height and fold-away).
CAUTION: Do not use the footrest as a step. If the footrest is used as a step, the chair might tip over.
 - Casters (wheel locks).
CAUTION: To ensure the subject is safely seated, ensure the chair wheels are locked before seating the subject.
1. Confirm that the chair wheels are locked.
 2. Seat the subject.
If performing an assisted chair-to-chair transfer with the Deluxe Chair, the operator may lower one of the armrests first.
 3. Adjust the Deluxe Chair for comfort.
 - a. If necessary, adjust the height and angle of the backrest.
 - b. If necessary, adjust the armrests.
 - c. If necessary, adjust the footrest.The subject should be seated comfortably in an upright position.

6.5 Fit the Subject Harness and Hip Belt

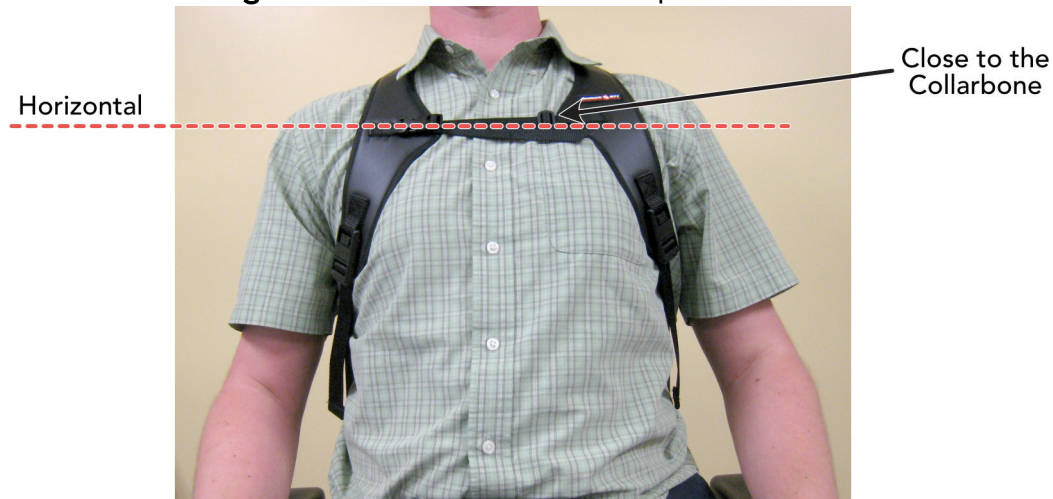
The Subject Harness and Hip-Belt on the optional Deluxe Chair are intended to be used to provide postural support for subjects with low trunk stability. They can also be used as a positioning aid or to assist subjects to maintain a consistent posture. It is not necessary to use these features with subjects who have normal trunk stability.

The ideal location for the harness brackets is at or just above the top of the shoulders (see [Figure: 6.3](#)). However, for use with a Kinarm Lab, the height is not a critical adjustment and if they are within 5-10 cm of the top of the shoulders, then the harness should still provide sufficient support. The location of the brackets can be adjusted using two methods: adjusting the height of the backrest on which the brackets are mounted, or adjusting the location of the brackets on the backrest. It is much simpler to adjust the height of the backrest, so it is recommended that the brackets can typically be left in a mid-to upper-range position relative to the chair backrest for most subjects. In addition, the angle of the brackets on the backrest may be adjusted if necessary. If the subject experiences discomfort or if the harness does not provide sufficient stability, adjust the harness and belts as needed. See the Kinarm Lab Hardware Guide for more details on the adjusting the Deluxe Chair.

Figure 6.3: Harness Bracket Height



The ideal location of the horizontal chest strap is just under, but not on, the collarbone, as shown in [Figure: 6.4](#).

Figure 6.4: Harness Chest Strap Location

1. Fit the hip-belt to the subject.
 - a. Loosely connect the belt together using the Velcro attachment.
If there is not enough length, or if there is too much overlap, adjust the belt length at both side clasps
 - b. Clip the belt clip together and pull the strap until it is snug.
2. Adjust the harness brackets, if necessary.
 - a. Raise the backrest first, and then if that is not sufficient, adjust the location and/or angle of the brackets on the backrest.
The top of the harness brackets should be near the shoulder (see [Figure: 6.3](#))
3. Attach the harness.
 - a. Place the harness on the subject, one side at a time.
It may be helpful to unclip the side clips, but if the straps are loose enough, it will typically be easy to pass the subject's arm through the harness without releasing the side clip.
 - b. Clip the chest clip in place.
4. Adjust the length of the shoulder straps.
The chest strap should run horizontally across the chest just under, not on, the collarbone (see [Figure: 6.4](#)).
5. Adjust chest strap length.
The chest strap should be snug without pressing the shoulder straps/padding into the subject's neck.
6. Tighten the side straps until the harness is snug against the subject.
7. Double-check all adjustments.
The subject should be secure and comfortable and the subject's arms should be able to move freely.

6.6 Position the Subject for the Subject Display

Positioning of the subject relative to the robots and subject display is important in order to ensure that the subject can correctly grasp the handle while both seeing and reaching as much of the workspace as possible. In addition, for some KST (e.g. Arm Position Matching), it is important to get the subject centered left-to-right within ~1 cm of midline. Positioning and setting up the subject at the subject display involves three aspects: vertical height adjustment, horizontal location, and use of the vision-blocking bib.

CAUTION: When the Deluxe Chair's footrest is folded up, ensure that the subject's heels are not underneath the footrest prior to lowering the chair, otherwise the subject's heels could be injured.

The subject should sit as high as possible with their legs still comfortably under the tabletop ([Figure: 6.5](#)). Having the chair adjusted as high as possible avoids having the subject hit or rest their arms on the tabletop and provides the best view of the subject display.

CAUTION: It is important to pay close attention to the position of the subject relative to the table top when adjusting the seat height. Raising the chair too high could compress the subjects legs, and arms and/or hands could be pinched against the table top during the adjustment.

CAUTION: The height of the subject is limited by the table-top. Because the table-top is present to safely position the subject outside of the robot workspace, it should not be removed.

Figure 6.5: Adjust the Chair Height



If using the Vertical Display, then the subject should be carefully pushed as close as possible to the table-top. (See [Figure: 6.5](#)). This position allows the subject to comfortably reach within the robot workspace. Make sure that the chair is approximately centred and directly facing the robot.

If using the Virtual Reality Display, then the subject should be carefully pushed close to the stand, with the subject sitting in an upright position (i.e. not slouching forward with their upper body) and their head tilted forward slightly. This position will allow them to view as much of the workspace as possible. If they are tall enough, it will also allow them to rest their head on the Virtual Reality Stand, see [Figure: 6.6](#)). Take care to avoid bumping the subject's head against the stand. Make sure that the chair is approximately centred left-to-right (i.e. their head should be aligned with the forehead cushion, so that they are no more than 1 cm off midline) and directly facing the robot.

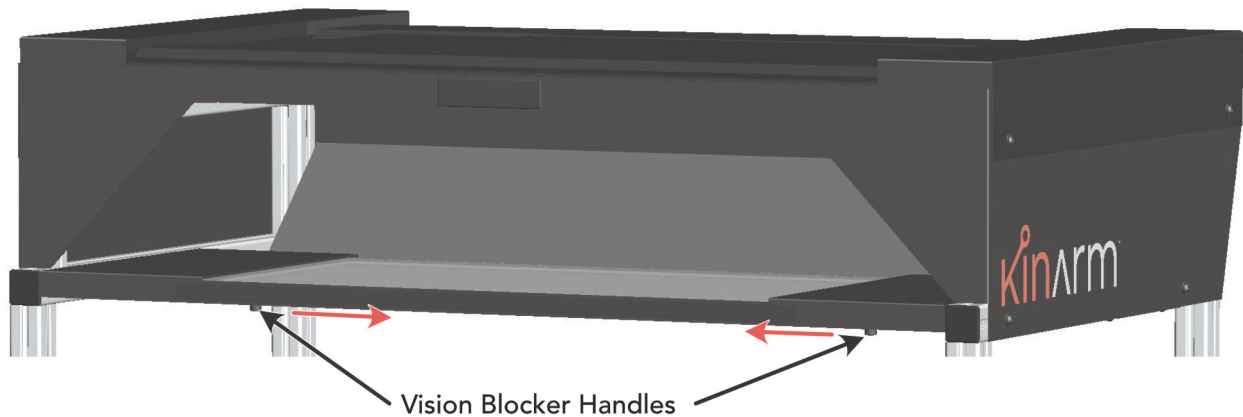
CAUTION: Pay close attention to the subject's knees, arms and head when wheeling up against the Table Top and the Virtual Reality Stand to avoid bumping them.

NOTE: With small children, they may find it easier to kneel on the chair for better positioning.

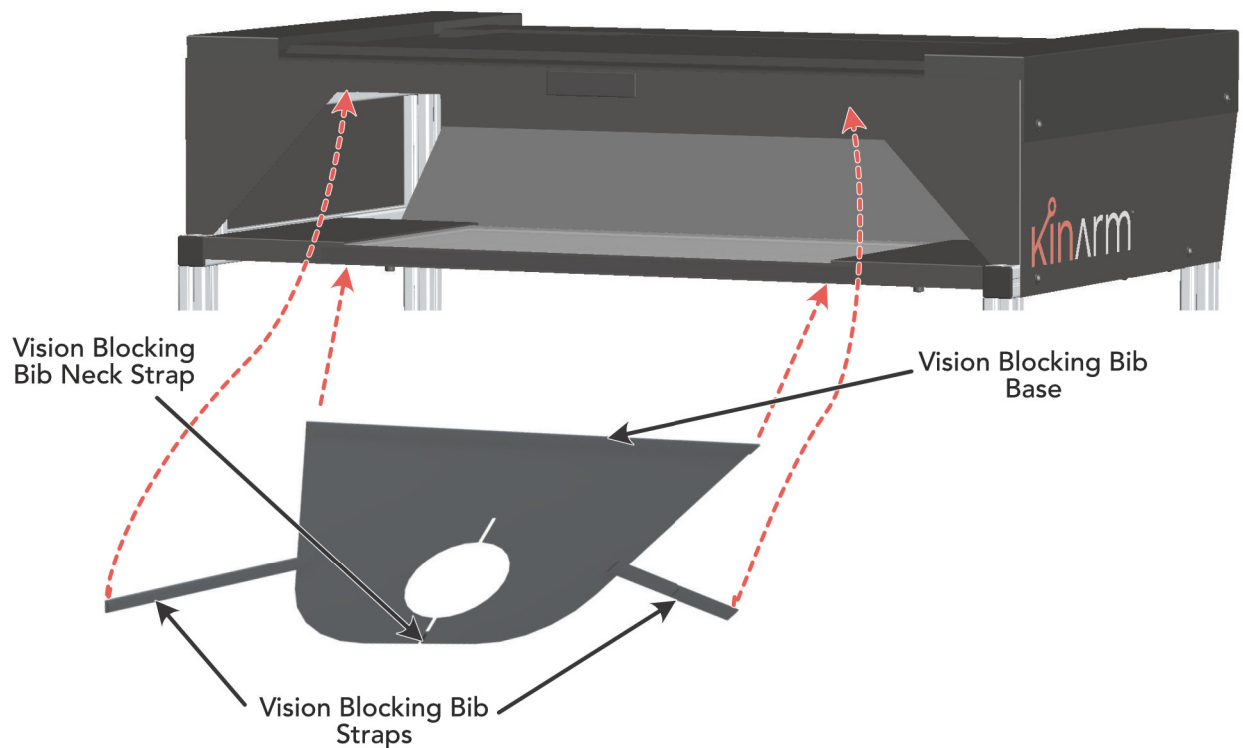
Figure 6.6: Position the Subject for Virtual Reality Display



As part of the vision-blocking system in the Virtual Reality stand, there are vision blocker panels that slide beneath the semi-transparent mirror to either block vision of the hands and robot (i.e. for Virtual Reality) or to allow vision of the hands and robot (i.e. for Augmented Reality). See [Figure: 6.7](#).

Figure 6.7: Vision Blocker Panel Setup

The function of the bib is to block vision of the arms, without the bib rubbing on the arms during movement. It is typically used in conjunction with the vision blockers. It attaches to the Virtual Reality Display as shown in [Figure: 6.8](#), and includes a neck strap to clip around the back of the subject's neck. When using the bib, care should be taken to ensure that the straps do not impede motion or touch the arms.

Figure 6.8: Set Up Vision-Blocking Bib

1. If using the Deluxe Chair, ensure that the subject's feet are on the footrest.
2. Unlock the wheel brakes on the subject chair.

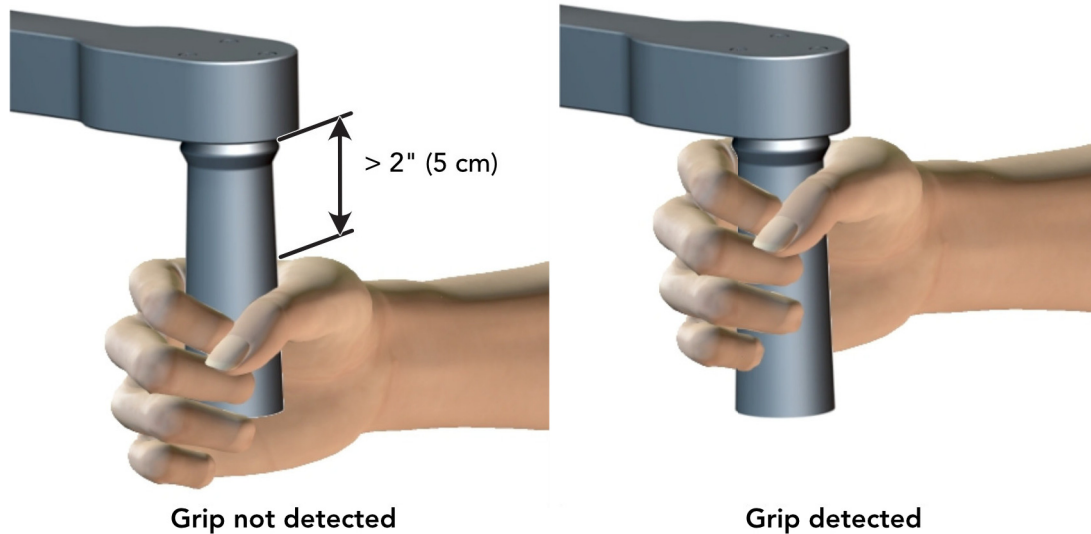
3. Roll the subject up to the system.
The subject should be centered (left-to-right) and close to the robots.
4. Adjust the seat height.
 - Standard Chair setup tips:
 - Lower the chair height as needed using the chair's built-in height adjustment handles until the fit is comfortable.
 - Deluxe Chair setup tips:
 - Fold away both armrests to prevent interference with the table top during setup and while running tasks.
 - Use the push bar to make adjustments to the position of the chair as needed.
 - Adjust the height of the seat as needed using the wired remote control.
5. Set the brakes on the chair.
 - Standard Chair: lock the three rear wheels.
 - Deluxe Chair: lock the foot pedal brake.
6. Attach the vision-blocking bib.
 - a. Place the vision-blocking bib gently around the subject's neck, under their chin and attach the straps to the underside of the subject display support on the Virtual Reality Display.
 - b. Close the neck strap around the back of the subject's neck.
 - c. Adjust the straps for optimum fit.
Take care to ensure that the straps do not impede motion or touch the arms.
7. Slide the vision blocker panels from both sides to the centre to block the view of the subject's arms and hands through the semi-transparent mirror (see [Figure: 6.7](#)).
The subject's ability to see through the mirror should be completely blocked.

6.7 Describe Handle Operation to Subject

Sensors built into each Kinarm End-Point Robot detect if the handle is being properly grasped. The robots will prevent movement during a task if the grip sensor detects that the handle is not being grasped properly. This functionality is achieved through "active braking". Similar to physical brakes, active braking stops the movement by zeroing all nominal task forces while simultaneously applying a highly viscous force field.

NOTE: If no Task Program is running, the motors are disabled. The robot, at this point, is free to move regardless of whether the handle is grasped properly or not.

The grip sensors trigger when the subject's hand grasps the handle with a grip that is less than ~2" (50 mm) below the underside of the link ([Figure: 6.9](#)).

Figure 6.9: Grip Detection

NOTE: If a handle with wrist support is used, the subject will typically rest their hand on the support flange at the bottom of the handle. This means that the hand may not be at the top of the handle, however, this is usually not a problem because the grip sensors have sufficient range to detect a hand resting on the support.

1. Instruct the subject to grab the robot handle as high on the handle as possible without touching the linkage.
2. While watching the grip status icons in the Status bar in Dexter-E (see [Figure: 4.1](#)), instruct the subject to slide their hand down the handle until grip is no longer detected, to ensure that the subject understands what not to do during the tasks.

6.8 Fit the Optional Handle Strap

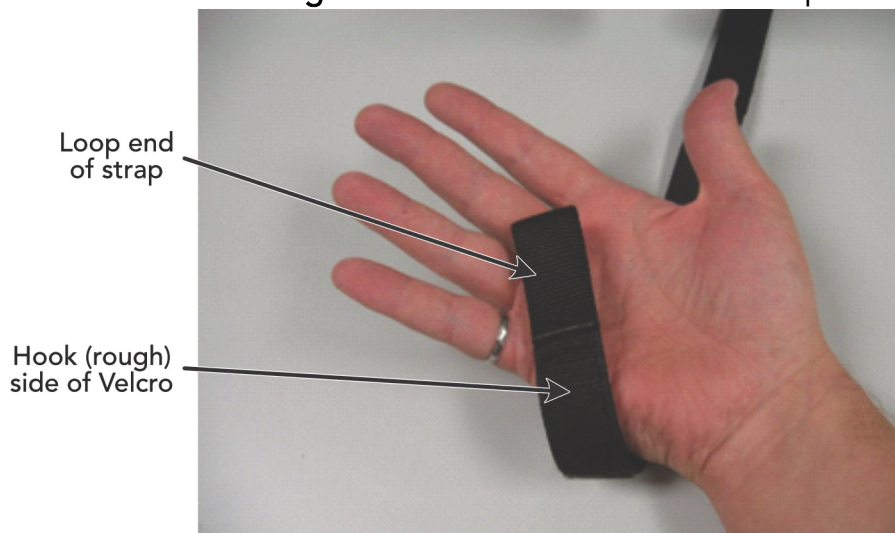
The handle strap is intended to keep a subject's hand on the handle of the robot during the session. This is useful for subjects who are not able to grasp well enough or for long enough to complete a task. The strap can be used with either the stand handle, or the optional handle with wrist support. The goal is to ensure that the strap is tight enough to maintain proper grip, but not so tight that it is uncomfortable.

The following instructions are written for setting up the strap for the right hand. Follow the mirror image setup for setting up the left hand.

CAUTION: When the subject's hand is attached to the robot handle with the handle strap, the subject will not be able to let go of the robot handle. The emergency stop button should always be available to the operator so that the robots can be stopped if needed.

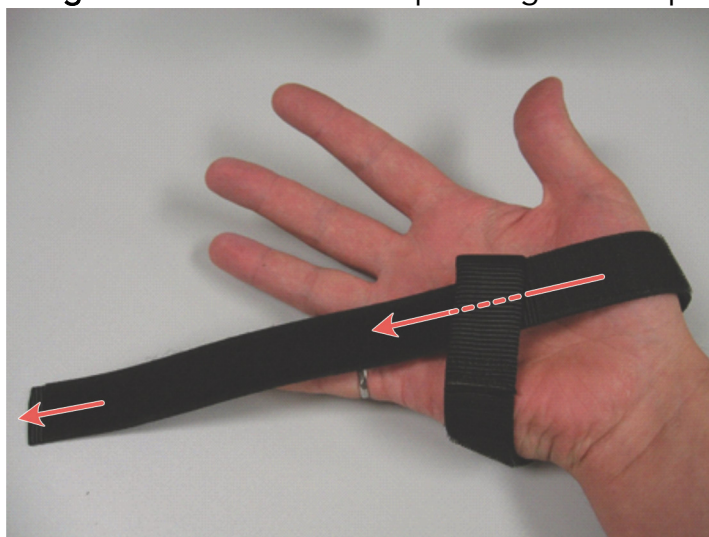
1. Position the strap (see [Figure: 6.10](#)).
 - a. Lay the strap on a flat surface (e.g. the table top) with the Velcro side DOWN, and with the strap loop towards the subject's torso.
 - b. Put the subject's hand over top of the strap with palm UP and fold the loop side of the strap onto the palm of the hand.

Figure 6.10: Position the handle strap



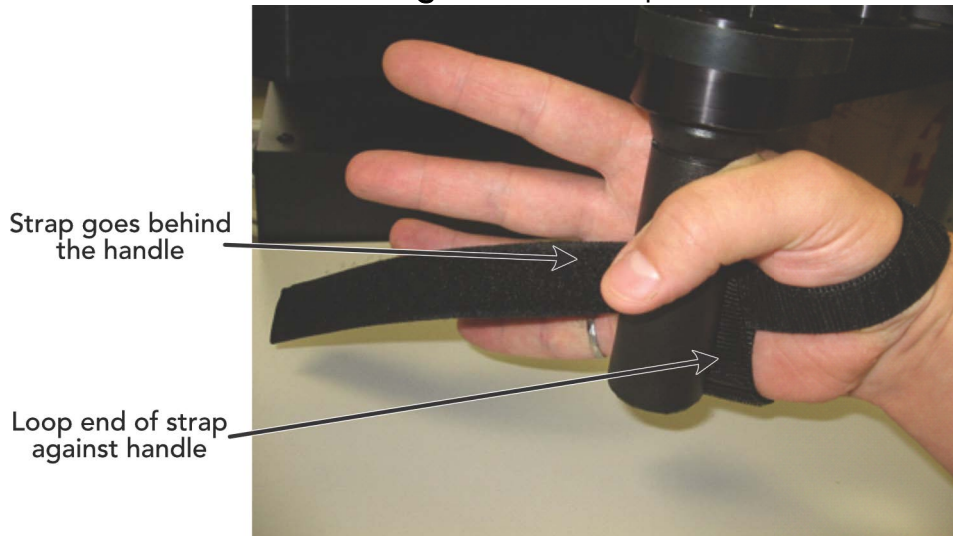
2. Loop the strap around the base of the palm (see [Figure: 6.11](#)).
 - a. Bring the non-loop end of the strap over the hand and feed it through the loop.
 - b. Pull the strap so that it is snug around the base of the palm (not the wrist itself), and so that the loop end of the strap is centred in the palm.

Figure 6.11: Feed the strap through the loop



3. Position the hand on the robot handle (see [Figure: 6.12](#)).
Help the subject grasp the handle as usual, making sure that the strap is between the hand and the handle.

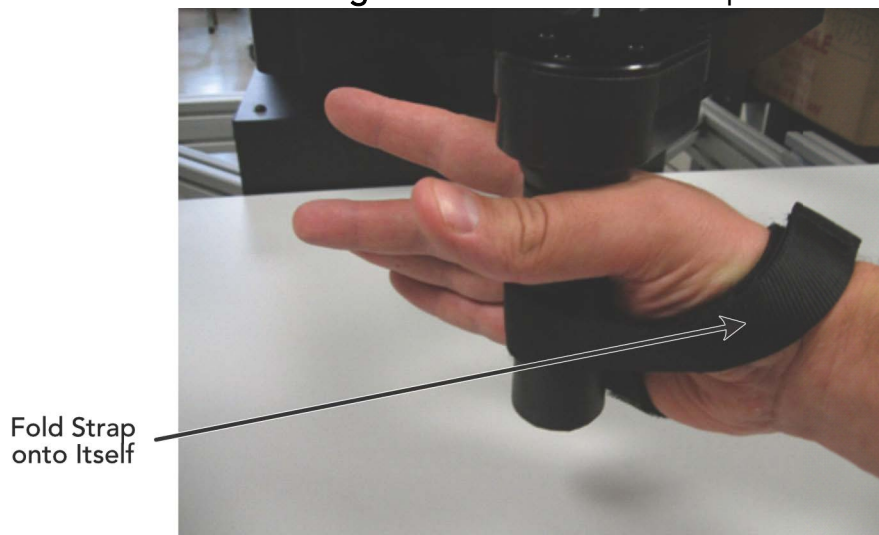
Figure 6.12: Grasp the handle



The loop end of the strap should be against the handle.

4. Secure the subject's hand to the robot handle (see [Figure: 6.13](#)).
 - a. Fold the strap around the robot handle and back onto itself.
 - b. Pull the strap until the subject's hand is comfortably against the handle.

Figure 6.13: Secure the strap



When properly set up, the subject will be able to release their grasp of the handle itself and the strap will keep their hand in place on the robot handle.

7 Calibration

For a Kinarm Lab to correctly measure position and force data and to apply the correct forces, the robots must be fully calibrated. If the robot or various peripherals are not calibrated, the collected data may be incorrect, the visual stimuli may not be aligned properly with the subject/robot and inappropriate forces could be applied by the robot to the subject.

The Kinarm End-Point lab, PN 14332 includes absolute encoders so that the robot is always automatically calibrated. The operator therefore only needs to calibrate any of the optional peripherals.

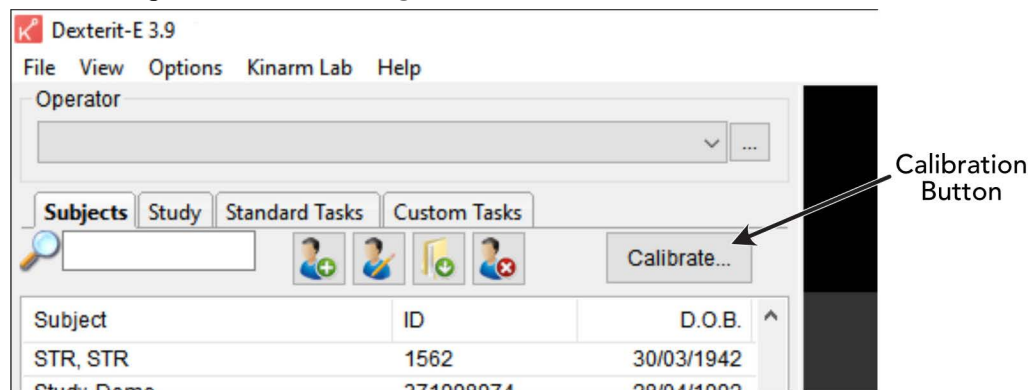
7.1 Zero the Force/Torque Sensors

If the lab is configured with force/torque sensors, then the sensors must be zeroed as per the following instructions. The *Calibration* dialog indicates with large icons whether or not a particular calibration step is required, and what its status is. If a green check mark appears beside a calibration step, that calibration step has been completed and does need not be repeated.

ATTENTION: It is important to ensure that the handle(s) remain free of any grasp or force during the zeroing process. If there is force applied, the force/torque sensor will not be properly zeroed.

1. Ensure that a subject has been selected.
2. Click **Calibrate**.

Figure 7.1: Initiating Calibration in Dexter-E



The Calibration dialog opens indicating which calibration steps are required.

Figure 7.2: Calibration Dialog

Kinarm End-Point Calibration

Calibration status

The robots are calibrated automatically at power on.

Left Kinarm: Right Kinarm:

Force sensor bias

While allowing the handles of all Kinarm Robots present to sit freely, click "Reset zero" to reset the bias vectors of all force sensors.

Reset zero

Left arm absolute force (N) Right arm absolute force (N)

Kinarm Gaze-Tracker calibration

Calibrate Kinarm Gaze-Tracker

Click Done to stop calibration at any time.

Done

3. Release the handles and then click **Reset zero**.

The values reported in the *Calibration* dialog for the force/torque sensors are the absolute values of the force in the X/Y plane. They are always positive.

7.2 Kinarm Gaze-Tracker Calibration

To ensure valid data with the Kinarm Gaze-Tracker, each subject must be properly setup and individually calibrated. The initial setup was described previously:

- Makeup and obstruction removal, and target placement (as per [6.3 Prepare the Subject for the Kinarm Gaze-Tracker](#))

The next steps are described in this section:

- [7.2.1 Adjust the Subject's Head Position and Orientation](#)
- [7.2.2 Confirm that Gaze is Trackable](#)
- [7.2.3 Calibrate and Validate the Gaze Tracking](#)

There is some freedom for the subject to move their head during these steps (and during data collection), but movement should be kept to a minimum to ensure that the target sticker always remains visible.

ATTENTION: Do not alter the position or orientation of either the camera or illuminator of the optional Gaze-Tracker. These components are carefully set during initial installation and should not be changed.

ATTENTION: Do not alter the settings on the optional Gaze-Tracker computer; manual changes may prevent gaze tracking from working correctly.

7.2.1 Adjust the Subject's Head Position and Orientation

Perform this task only after the subject has been seated and the robot(s) calibrated to the subject.

When first sitting in a Kinarm Lab, subjects have a tendency to hold their head vertical and look horizontally towards the back wall of the Kinarm Virtual Reality display. Not only will such a posture be uncomfortable for the subject during a long task, but it will often prevent Gaze-Tracking from functioning properly.

Tilting the subject's head accomplishes the following objectives:

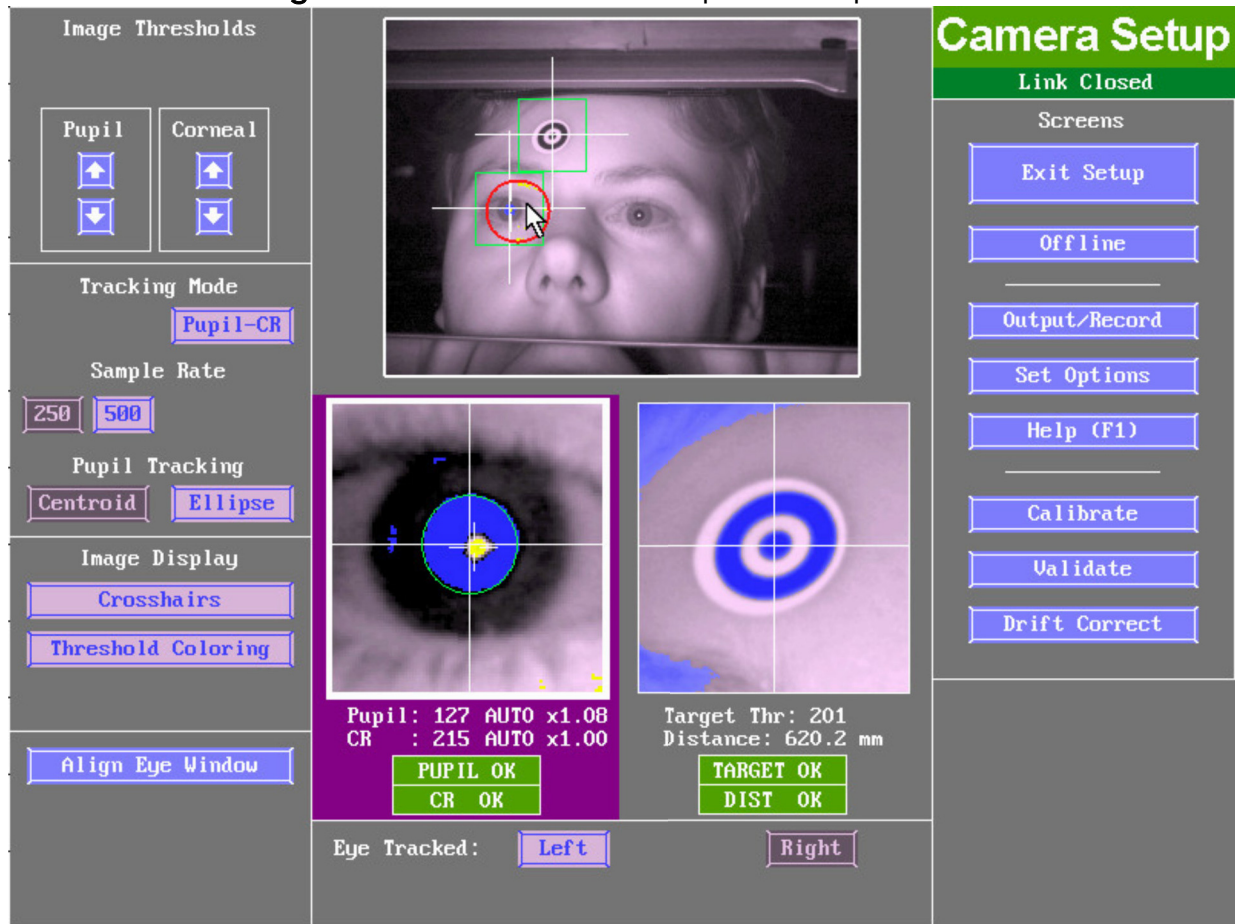
- Tilting ensures that the subject will be able to see targets that are proximal to their body. This can be tested by having them move the Kinarm Robots towards their body and asking the subject to look at their hands while they do this.
- Tilting minimizes the odds that the target sticker for gaze tracking will be at too oblique an angle to be detected properly.
- Tilting helps prevent the lower eyelid from blocking the camera's view of the subject's eye. If the lower eyelid occludes the pupil or is close to occluding the pupil, consider having the subject increase the tilt of their head.

In addition to controlling head tilt, the position in the workspace needs to be controlled as well. Positioning a subject's head accomplishes the following objectives:

- If the subject's head position is too far off mid-line, then their eye or the target sticker may not be in the camera's view and gaze tracking will fail.
 - Although the Kinarm Gaze-Tracker can tolerate head movement, the range of movement is limited by the viewing angle of the camera. Head movement should be limited as much as possible.
- It is important to locate their head vertically in the correct location so that the camera can view both the eye and the target sticker simultaneously.
 - If the Gaze-Tracker Computer cannot view the eye or target sticker, then gaze tracking will fail.

Head position and tilt can be viewed using the Gaze-Tracker Computer setup screen (see [Figure: 7.3](#)).

Figure 7.3: Gaze-Tracker Computer Setup Screen



Check Subject's Head and Body Tilt

1. Have the subject adjust their posture using the following guidelines:
 - a. While grasping the Kinarm robots, the subject's hands should be near the middle of the workspace.
 - b. When looking at their hands, the subject's head should have a 20-30° tilt relative to vertical.

ATTENTION: As the tilt angle of the subject's head increases, subjects will likely find it more comfortable if some of the tilt is achieved by curving their upper back and not just angling their neck. Actual head tilt is best observed by looking at the subject's profile from the side. Experienced Kinarm Lab operators will be able to tell by viewing the subject's face on the *Camera Setup* screen.

2. Determine if the lower eyelid is occluding the pupil.
Look at the subject's face on the *Camera Setup* screen. If the lower eyelid is partially occluding the pupil, then the subject should tilt their head further

Check Subject's Head Position

3. Confirm that the subject's head position is suitable both horizontally (left-to-right) and vertically.

Suitable head positioning is determined by looking at the subject's face on the *Camera Setup* screen.

If the subject positioning instructions for the robot are followed correctly (i.e. the subject is centred on the virtual reality stand with their head on the headrest), then the subject should typically already be in the correct location. If their head position is not correct, consider adjusting their body position.

Check Position of the Target Sticker

4. Look at the location of the target sticker relative to the viewable region.

If the target is not visible, or it is very close to not being visible, adjust the target sticker or the subject's vertical position relative the Kinarm Virtual Reality stand.

5. In the *Camera Setup* screen, click on the pupil eye that is to be tracked.

If there are red boxes on the image (one around the target and one around the pupil) and there is a NEAR EYE warning on the *Camera Setup* Screen beneath the magnified image of the target, then the target sticker is too close to the eye in the vertical direction (see [Table: 8.1](#)).

6. Verify that there is a set of cross-hairs on the magnified image of the target sticker.

If not, then the Gaze-Tracker Computer cannot detect the target sticker. See [Table: 8.1](#) for common problems and trouble-shooting tips.

7.2.2 Confirm that Gaze is Trackable

Prior to calibration, some simple tests can be performed to confirm that gaze will be trackable over the workspace. If a subject has a 'dominant' eye that they are aware of, it is typically best to attempt to track that eye.

Typically the camera focus does not need to be changed from one subject to the next. However, if it is deemed necessary, optimal focus is determined by making the corneal reflection (CR) as crisp and clear as possible (see image of pupil in [Figure: 7.4](#), and review the EyeLink User Guide for more details).

If the target is not detected, then the CR and pupil will not be detected, so issues with target detection must be resolved before proceeding. Likewise, if something is interfering with CR detection, then the pupil will not be detected, so issues with the CR need to be resolved before proceeding. If the pupil is not detected, then gaze tracking will not occur. Various trouble-shooting tips for each of these situations are summarized in [Table: 8.1](#).

If there are problems detecting gaze reliably over the nominal calibration region it is possible to adjust the size of the calibration region to make it smaller (see [Figure: 7.5](#)). However, using a smaller calibration region can impact the quality of gaze-tracking data near edges of the trackable region.

Figure 7.4: Gaze-Tracker Computer Camera Setup Screen

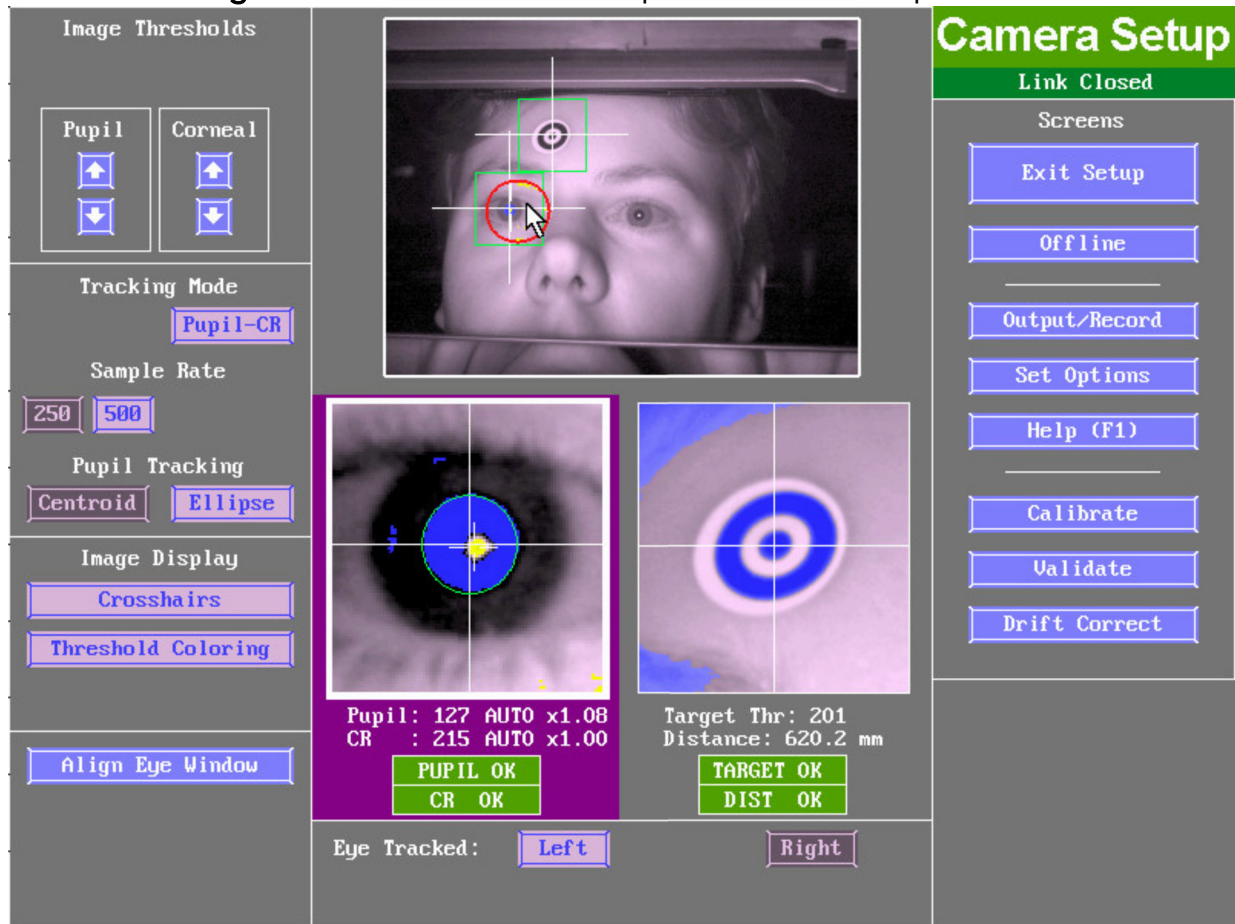
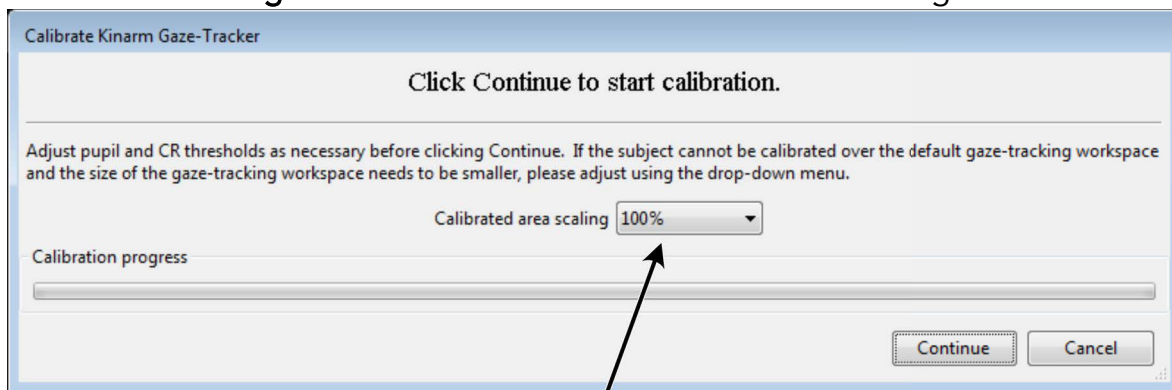


Figure 7.5: Calibrate Kinarm Gaze-Tracker Dialog



Scaling Factor

1. In Dexterit-E, click **Calibrate** then click **Calibrate Kinarm Gaze-Tracker**
Dexterit-E will display a number of targets on the subject display at the centre and outer limits of the region where gaze tracking is possible.

2. Click on the pupil of the eye to be used for tracking.

NOTE: Regardless of which eye is actually being tracked, the “left” text box below the magnified view of the subject’s eye on Camera Setup screen will be selected.

A red circle indicates which eye is selected for tracking ([Figure: 7.4](#)).

3. Confirm that the camera’s focus is correct.
 - a. Instruct the subject to look at the centre target, while viewing the image of the pupil. The CR should be small and crisp.
 - b. If necessary, adjust the camera focus and repeat.
4. Confirm that gaze is trackable throughout the workspace.
 - a. Instruct the subject to look sequentially at each of the outer targets while viewing the image of the pupil, repeating as necessary
During this stage, instruct the subject to try to keep their eyes as wide open as comfort allows to improve the tracking.
The gaze-tracker should successfully find the target, CR and pupil at each of the targets.
 - b. If necessary, adjust pupil and CR thresholds.
In general, after threshold adjustment, pupil thresholds should be between 75 and 115 and CR thresholds should not exceed 240. Refer to the EyeLink User Guide provided with your Kinarm Lab, in particular the section on “Setting Pupil Thresholds”, for tips on how to set the thresholds correctly.
 - c. If necessary, select a smaller scaling factor for the calibrated gaze region (100%, 90%, 80%, 70%) in the *Calibrate gaze-tracking* dialog, and repeat the previous steps.
Some subjects will not be trackable over the nominal region, and therefore a smaller calibration region will need to be used.

7.2.3 Calibrate and Validate the Gaze Tracking

The calibration and validation steps for gaze-tracking are very similar to each other. The subject will be presented with a series of targets to look at. From a subject’s perspective, it will be appear as one long sequence, even though the same targets are shown twice: once for calibration and then once for validation. The end result will be a quantitative assessment of the calibration accuracy, shown in the *Validate calibration* dialog (see [Figure: 7.6](#)). The results will be one of the following:

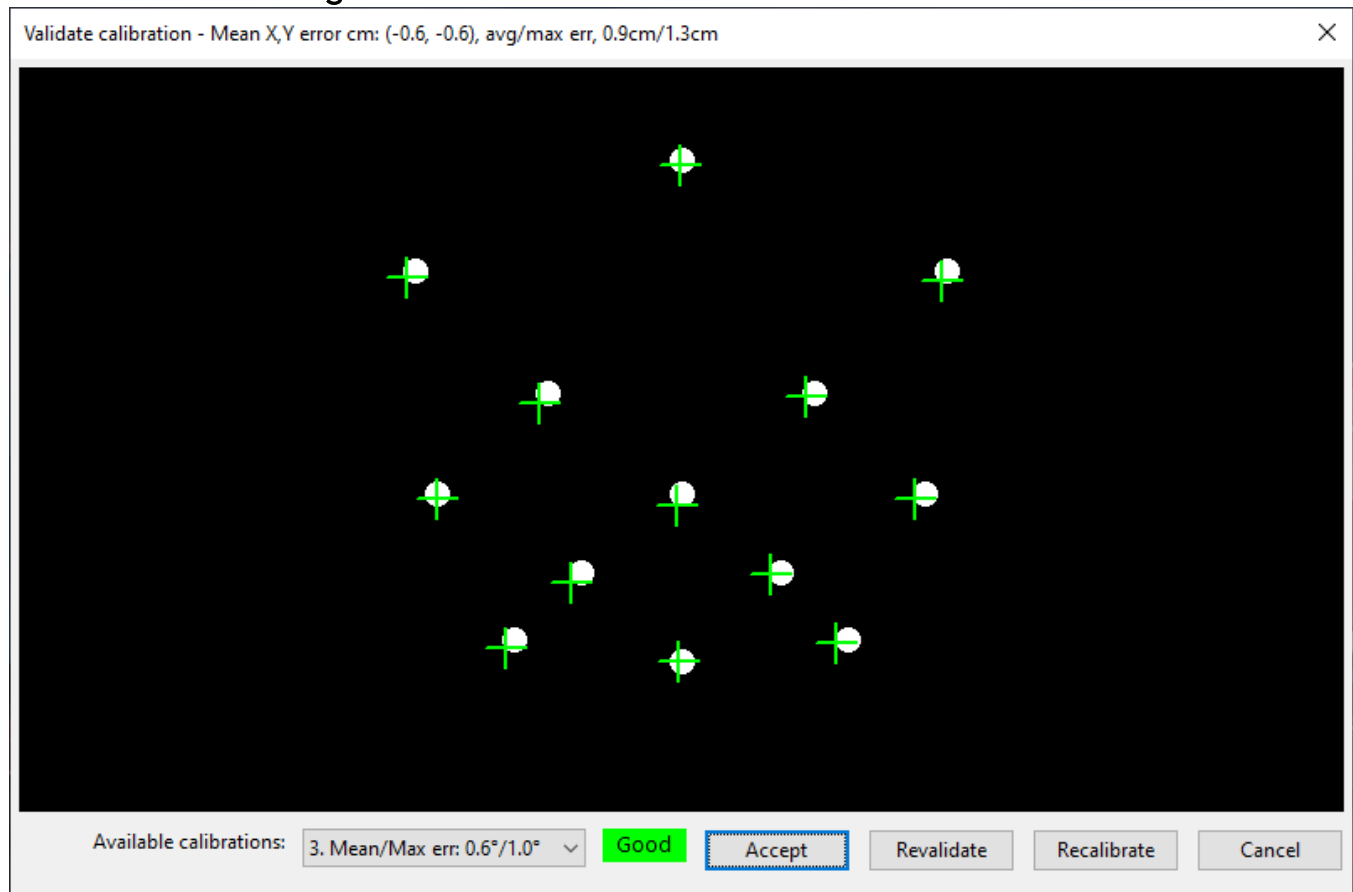
- Good: mean error < 1.0° AND maximum error < 1.5°
- Poor: mean error > 1.5° OR maximum error > 2.0°
- Fair: all other error values.

While a Good calibration is always acceptable, there are situations in which a Fair or Poor calibration may be acceptable as well. For example, if timing of eye movements is the only critical measure for a given experiment, and not accuracy of gaze direction, then a Good

calibration may not be necessary. More generally, if the results are Fair, then it is recommended to click **Recalibrate** to try to get a Good calibration, however, if there is not sufficient time to repeat the calibration process, or if several calibration attempts have already been tried, then a Fair calibration will likely be acceptable. With a Poor calibration result, typically it is recommended to click **Recalibrate**. However, depending on where in the trackable region the errors occurred (i.e. in relation to the task that is going to be performed), a Poor calibration might be considered acceptable.

Dexterit-E can be configured to use either 5 or 13 (legacy) points for calibration. In general we recommend using the 5-point calibration because it is much faster and with very little loss in accuracy. See the Dexterit-E User Guide for further details on changing to 13 points if desired.

Figure 7.6: Gaze-Tracker Calibration Results Window



1. Instruct the subject on how to perform the calibration and validation:
 - a. Instruct the subject that they will be presented with a sequence of targets across the workspace.
 - b. Instruct them to look at the CENTRE of each target and to CONTINUE looking at the centre of each target until a new target appears.

2. Click **Continue** on the *Calibrate Kinarm Gaze-Tracker* dialog to begin.

See [Figure: 7.5](#).

The sequence of targets will proceed automatically as they look to each of the targets. A validation sequence will automatically follow the calibration sequence.

3. Monitor the image of the subject's eye and target on the Gaze-Tracker Computer's screen as the calibration/validation sequence progresses.

If either the pupil or target is lost (i.e. cross-hairs disappear), or if the tracking of one seems inconsistent (e.g. the ellipse around the pupil is jittery in size/shape), then you will likely need to stop the calibration, re-adjust thresholds or deal with occlusion and re-start the calibration.

If the calibration sequence appears to be "stuck" and is not automatically advancing, check to see if there are any errors on the Gaze-Tracker Computer's screen. If not, then click **Auto-Trigger** to manually accept the calibration point. Because that action will also switch the EyeLink to "manual" mode, you will then need to click **Auto-Trigger** again on the Gaze-Tracker Computer to ensure that the process remains automated.

For further tips on trouble-shooting the setup of a subject for gaze-tracking, see [8.4 Data Collection Using the Kinarm Gaze-Tracker](#).

Once the sequence has finished, a new window will display the results of the calibration/validation step, classified as either **Good**, **Fair**, or **Poor**. See [Figure: 7.6](#).

4. If the results are acceptable, then click **Accept** to finish the calibration.

5. If the results are not acceptable, then click **Recalibrate**.

When the recalibration finishes, the *Validation* dialog will now include a combo box that will allow you to select the current calibration or any of the previous calibration attempts since the *Calibration* dialog was opened.

8 Data Collection

The Dexterit-E software is used to run individual Kinarm Standard Tasks, Custom Tasks, or an ordered sequence of tasks specified in a Study Protocol.

CAUTION: Prior to running a task in labs with a vertical display, instruct bystanders to keep out of the robot workspace. As there is no protective barrier around the workspace, bystanders risk injury.

ATTENTION: Ensure there are no programs running simultaneously with Dexterit-E such as virus scanners, web browsers, copying files, e-mail, watching movies, etc. These programs can negatively affect data acquisition and timing on the subject display.

NOTE: Operators may choose to dim the room lights or close window blinds as needed to prevent bright lights from shining into the Virtual Reality Display. Reducing bright lights helps to provide more consistent visual presentation of targets during the tasks.

8.1 Select a Task from a Study

The active study defaults to the study that the active subject has been assigned to in the subject database record. If the active subject is not assigned to any study, then the task list will be empty. This default behaviour may be overridden by selecting any of the available studies from the Study drop-down list. If a different study is selected for the active subject, then the operator will be asked to confirm if the subject should be assigned to that study permanently (i.e. update the subject's database record with the new study), or if the assignment is only temporary. If temporary, only that session's data will be assigned to the new study.

The active session defaults to either the next available session in the active study (if the most recent session was completed) or the most recent session (if the most recent session was not completed). If the intended session to be performed does not correspond to a specific session that is already defined in the study then there is an "ad-hoc" option available in the list. You can collect multiple ad-hoc sessions as part of a study.

ATTENTION: The Study Protocol does not force the task order. The operator can skip tasks in the list and will not be warned that they have done so.

NOTE: Any task in the Study Protocol can be re-run. The status indicator will indicate the status of the last run.

1. Confirm the operator is selected from the list of available operators.
If the current operator is not present in the list, click ... to add the current operator to the list.
2. Click on the *Study* tab.

3. Confirm that the correct study is selected.
If not, then select a study from the list of available studies.
4. Confirm that the correct session is selected.
If not then select a session from the list of available sessions
5. Click on the first Task Protocol in the list and review any notes to the operator.
The note may state which arm is to be assessed.
6. Select whether the **Left** or **Right** arm is to be assessed.

ATTENTION: The operator should take care that the correct arm is assessed. Dexterit-E cannot detect if the selected arm does or does not match an instruction in the note.

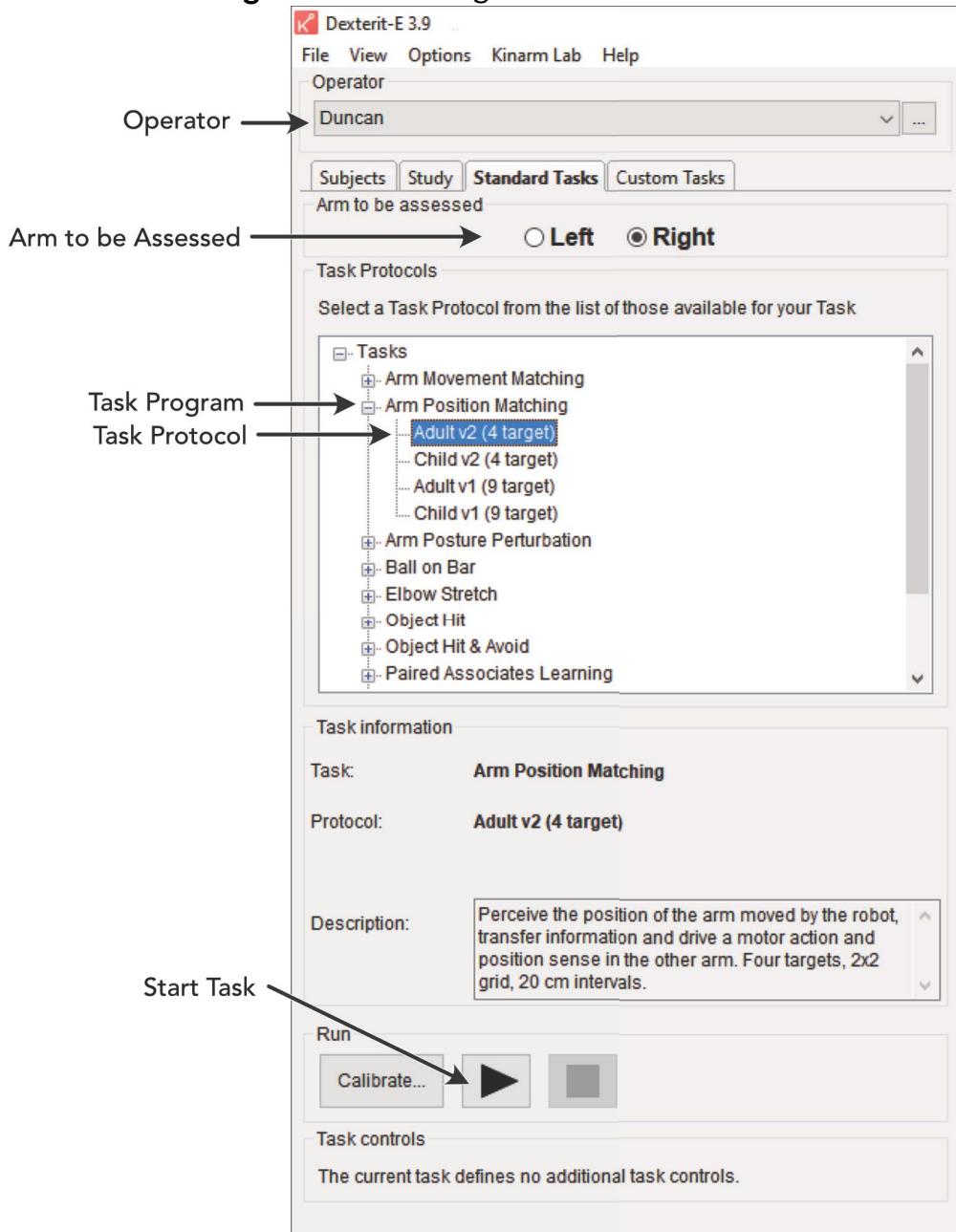
8.2 Select a Custom or Standard Task

When not using the *Study* tab, tasks can instead be selected and run from either the *Standard Tasks* tab or the *Custom Tasks* tabs ([Figure: 8.1](#)). The functionality of the two tabs is nearly identical. Kinarm Standard Tests are an optional module within Dexterit-E. Each test comprises a Standard Task and Standard Report.

NOTE: For some KST (Trail Making A, Trail Making B, and Object Hit and Avoid) there are many possible protocols that can be run. By default, if the subject has not performed one of these exams previously, then Dexterit-E will randomly select a protocol to use. If the subject has performed one of these exams previously, then Dexterit-E determines the last protocol used and selects the next available protocol to run. Right-clicking on the protocol name will show a context menu with the list of available protocols and a check mark will be beside the currently selected protocol. The context menu allows you to select a specific protocol if desired.

- For Trail Making, when Dexterit-E automatically selects a new protocol the result is that the pattern of targets will be different for each exam.
- For Object Hit and Avoid, when Dexterit-E automatically selects a new protocol a different set of target shapes will be selected for each exam.

Figure 8.1: Loading a Standard Task in Dexter-E



1. Select the **Standard Tasks** tab or **Custom Tasks** tab, as applicable.
2. Confirm the operator is selected from the list of available operators.
If the current operator is not present in the list, click ... to add the current operator to the list.
3. Select whether the **Left** or **Right** arm is to be assessed.
Task Programs are typically designed to be applicable to either the right or left arm (or in the case of a bilateral task, to be reversible from left-handed to right-handed).

4. Select a Task Program and Task Protocol from the alphabetic list.
Expand the list to show the Task Protocols associated with that Task Program.
5. Confirm that the vision blocking system (bib and panels) are properly set up for the tasks selected.

ATTENTION: The vision-blocking system must be used for KST such as the Arm Position Matching or Arm Movement Matching tasks to ensure the subject has no visual information on the location of their arms. For efficiency of time, it is recommended that the bib be worn and the vision blocker panels closed for the entire assessment.

8.3 Run a Task

Once a task has been selected on one of the tabs, the Operator can click the **Play** button to initiate a process that will start a task. If written or video instructions have been specified for the task, they will be displayed to the operator to allow them to instruct the subject, prior to the task starting.

After a task is started, the **Play** button changes to a **Pause** button. The effect of the **Pause** button depends upon the Pause/Unpause settings of the Task Protocol (see the Dexterit-E User Guide for descriptions of the various pause options available for Custom Tasks and which options have been selected for KST).

The **Stop** button also becomes activated once a task has started. Clicking **Stop** while a task is running or paused will prompt the operator to confirm if they want to stop the task and reset it back to its initial state.

ATTENTION: If the motors are disabled by the operator using the emergency stop button or because the handle is released by the subject during the task, then the task should be stopped and restarted.

ATTENTION: The subject must understand the instructions before a Standard Task is run. If they do not understand the instructions, then the task should not be run. It is recommended that the subjects be appropriately screened prior to running the tasks (see [6.1 Screen Subjects](#)).

ATTENTION: In rooms with lots of ambient light, it is recommended that overhead lights be blocked/dimmed. This will ensure that the subject can see properly in the Virtual Reality Display

1. Select a task using one of the following methods:
 - On the *Study* tab, see [8.1 Select a Task from a Study](#)
 - On the *Standard Tasks* tab or *Custom Tasks* tab, see [8.2 Select a Custom or Standard Task](#)
2. Click the **Play** button.
If there are any textual or video instructions defined with the task they will be presented before the task begins. Otherwise the task will begin playing.

3. If applicable, read or show the video instructions.
 - a. If a Kinarm Standard Task is selected, play the instructional video or read the instructions as written. Custom tasks may or may not have textual and/or video instructions.

NOTE: The preferred language for Kinarm Standard Task instructions can be selected by navigating to **Options > Instruction Languages**.

- b. Click **Continue** to start the task.
4. Monitor task progress.

The task will now proceed according to the specifications in the Task Protocol and the Task Program. The task will continue to run until all specified trials have occurred. The elapsed time and remaining time (if available) of the task is displayed to the operator in the Task Progress panel in the upper right portion of the operator screen ([Figure: 4.1](#)).
5. Do one of the following as appropriate:
 - Wait until task finishes.
 - Press the **Pause** button.
 - Press the **Stop** button.

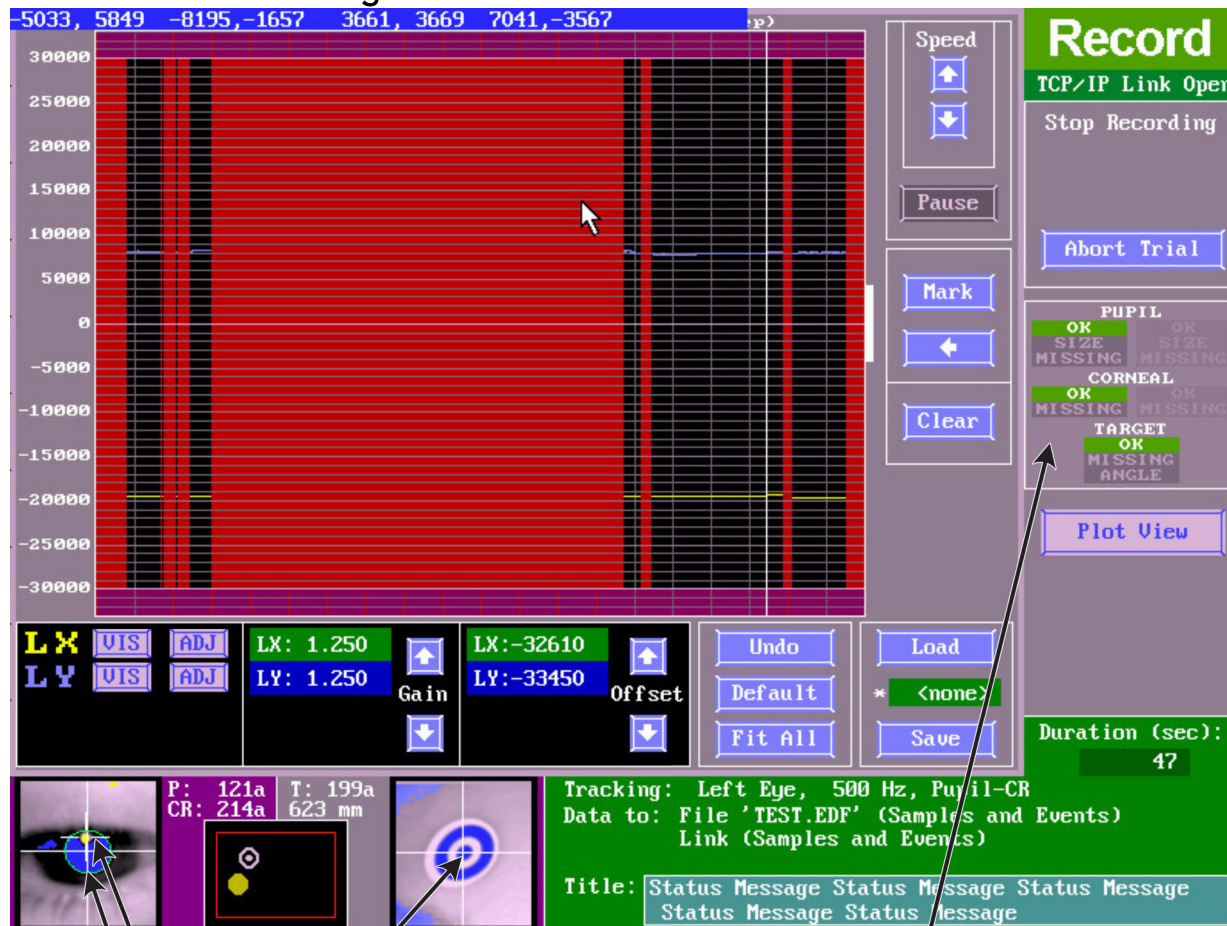
Once complete, the task will stop and Dexterit-E will wait for the next operator interaction.

8.4 Data Collection Using the Kinarm Gaze-Tracker

The Kinarm Gaze-Tracker must be monitored closely to ensure that the tracking has not failed. [Table: 8.1](#) provides some suggestions for common problems or errors that might occur while using the Gaze-Tracker.

When a task is started with Dexterit-E, the Gaze-Tracker Computer automatically switches to recording. Monitor the two images at the bottom: one for both the pupil and CR, and one for the target. There should be cross-hairs on each of the pupil, the CR and the target. If any or all are missing the cross-hairs, then gaze tracking has failed. See [Table: 8.1](#) for trouble-shooting tips.

Figure 8.2: Gaze-Tracker Record Screen



Cross-hairs should be visible

Status Indicators should all indicate "OK"

Table 8.1: Gaze-Tracker Troubleshooting Tips (Sheet 1 of 2)

Issue/Error displayed on Gaze-Tracker Computer Screen	Meaning/Description	What to Do Next?
Face not visible on Camera Setup screen.	Something is not setup correctly with the hardware.	• Check that the lens cap is off the camera. • Check that there is not a red LED lit up on the Gaze-Tracker power box (which indicates the illuminator fan is not running). • Check that the Gaze-Tracker power is on. • Check that there is nothing obstructing the camera's view of the subject.
Target: MISSING	The Gaze-Tracker cannot find the Target Sticker.	• If the sticker is not detected, can you see the sticker? If not then either move the sticker or move the subject's head. • Is something occluding the sticker (including stray hair)? If so, remove the occlusion.
Target: ANGLE	The angle of the Target Sticker is too oblique.	Reposition the Target Sticker more medially (if the large angle is left/right) or ask the subject to tilt their head down (if they are not tilted down) or reposition the Target Sticker on the subject's cheek.
Target: NEAR EYE or EYEDIST	The target is too close (vertically) to the pupil.	The Target Sticker needs to be moved up or down, further away from the pupil, and/or the subject may need to tilt their head more.
Target: DIST FAR	The subject's face is too far away from the camera.	Ask the subject to lean in closer to the Virtual Reality Stand.

Table 8.1: Gaze-Tracker Troubleshooting Tips (Sheet 2 of 2)

Issue/Error displayed on Gaze-Tracker Computer Screen	Meaning/Description	What to Do Next?
Corneal: MISSING or NO CR	The Gaze-Tracker cannot find the CR.	- Check for occlusions, such as hairs, eyelids, or eyeglasses. See 6.3 Prepare the Subject for the Kinarm Gaze-Tracker for information on dealing with occlusions. - Are other reflections (e.g. eyeglasses) creating a false CR? See 6.3 Prepare the Subject for the Kinarm Gaze-Tracker for dealing with eyeglasses. - Check the CR thresholds (i.e. is the CR yellow and does the yellow bleed outside of the CR?). See 7.2.2 Confirm that Gaze is Trackable for information on adjusting the thresholds.
Pupil: MISSING or NO PUPIL	The Gaze-Tracker cannot find the pupil.	- Check for occlusions, such as hairs, eyelids, or eyeglasses. See 6.3 Prepare the Subject for the Kinarm Gaze-Tracker for dealing with occlusions. - Check the pupil thresholds (i.e. is the pupil blue and does the blue bleed outside of the pupil?). See 7.2.2 Confirm that Gaze is Trackable for information on adjusting the thresholds.

8.4.1 Get Help from Kinarm Support

If you are having difficulties with gaze tracking, and need to contact Kinarm Support for help with the Kinarm Gaze-Tracker, please follow these steps.

1. Collect screenshots from Gaze-Tracker Computer
 - a. Click **Calibrate gaze tracking** on the *Calibration* dialog in Dexter-E ([Figure: 7.2](#)).
 - b. Have subject look at centre target, press **Alt+F7** on the Gaze-Tracker Computer. This captures a screenshot.

- c. While the subject looks at the visual targets in the four corners, capture screenshots. To capture the screen press **Alt+F7** on the Gaze-Tracker Computer. Make a note of how many screenshots are captured and in what order and what the subject was doing during each one.
 - d. Make a note of how many screenshots are captured and in what order and what the subject was doing during each one.
2. If the problem happened or is happening during a task, calibration etc:
 - a. Take a movie of the Gaze-Tracker Computer screen (i.e. with a camera or smartphone).
 - b. Make notes on what the subject was doing, what they were instructed to do, what types of problems you are observing.
3. Collect files from the Gaze-Tracker Computer.
 - a. Exit the Eyelink software to the File Manager by pressing **Ctrl+Alt+Q** on the Gaze-Tracker computer.
 - b. Navigate to the folder `\ELCL\EXE\`.
 - c. Within that folder, copy the most recent PCX files (these are the screenshots taken by pressing **Alt+F7**, one for each screenshot that you took) and the eye.log file.
4. Send eye.log, the relevant screenshots (i.e. PCX files) and, if you recorded a movie also include the movie, to Kinarm Support.

9 After the Tasks are Complete

Once the tasks are complete, the subject should be carefully removed from the robot and those areas contacted by the subject should be disinfected. Standard Reports can be generated and/or data analyzed.

9.1 Remove the Subject From the Lab

Reference the figures in [5 Lab Preparation](#) to assist with these steps.

CAUTION: Engage the wheel brakes during subject transfer. This precaution prevents the chair from rolling away during the transfer.

CAUTION: When transferring the subject, ensure that the raised armrest(s) are locked in the upright position.

CAUTION: Before transferring the subject, ensure that the Deluxe Chair's footrest is secured in the upright position. If the footrest is folded up, ensure that the subject's heels are not underneath the footrest prior to lowering the chair.

1. Remove the vision-blocking bib from around the subject's neck.
2. If applicable, remove the handle strap from the subject's hands.
3. Unlock the wheels/brake of the chair.
4. Wheel the chair away from the Virtual Reality Stand.
5. With the chair in the desired location, engage the wheel brake(s).
6. If necessary, ensure that the seat is lowered to an appropriate height for the subject so that they can more easily get out of the chair.
7. If a Deluxe Chair was used:
 - a. Fold up the armrests.
If the subject needs assistance with transfer from the chair, the armrest on the side from which the transfer will be made may be left folded down.
 - b. Fold up the footrest.
 - c. If applicable, unclip the harness and the hip-belt.
8. If the Kinarm Gaze-Tracker was used, remove the target sticker from the subject's face.
9. Instruct and/or assist the subject to stand up/transfer out of the chair.
Guide them away from the Kinarm Lab.

9.2 Clean the Lab

The Kinarm Lab should be cleaned between uses. Unless otherwise specified, suitable cleaning products include soap and water and suitable disinfectants which include mild bleach-based cleaners (e.g. Clorox wipes) or other disinfectant wipes (e.g. Lysol wipes).

The following items require special cleaning care:

- Semi-transparent mirror
 - Only use ammonia-based household glass cleaners (e.g. Windex) or 70% isopropyl alcohol.
 - Only use clean microfibre cloth for cleaning the mirror surfaces. Dirty cloths, brown paper towels, newspaper, bristled brushes or abrasive cleaners will damage the reflecting surface.
- Subject display
 - As needed, gently dust the screen of the subject display using a dry microfibre cloth. If there are fingerprints on the screen, a small amount of water can be used with the microfibre cloth.
 - To avoid damaging the subject display, do not apply pressure while wiping.
- Vision-blocking bib and forehead rest
 - These items can be disinfected with up to 20% bleach and water solution, hydrogen peroxide based cleaners or quaternary based cleaners.
 - Isopropyl alcohol may be used for stubborn stains.
 - Rinse thoroughly with water and air-dry.
- Harness and belt
 - The harness and belt must be hung to dry. Do not put them in a dryer.

CAUTION: All components and surfaces that came into contact with the subject should be cleaned between each use to prevent the transfer of pathogens.

ATTENTION: Unless otherwise specified, no part of the Kinarm lab should be submerged or subjected to large amounts of liquid.

1. Wipe clean and disinfect all components and surfaces that came into contact with the subject between each use.

The parts that are likely to require cleaning and disinfecting include:

- Robot Handles
- Table Top
- Vision-blocking bib and forehead rest
- The chair: seat cushion, backrest, and armrests
- The subject harness and hip-belt

2. Wipe clean all other components and surfaces as necessary.

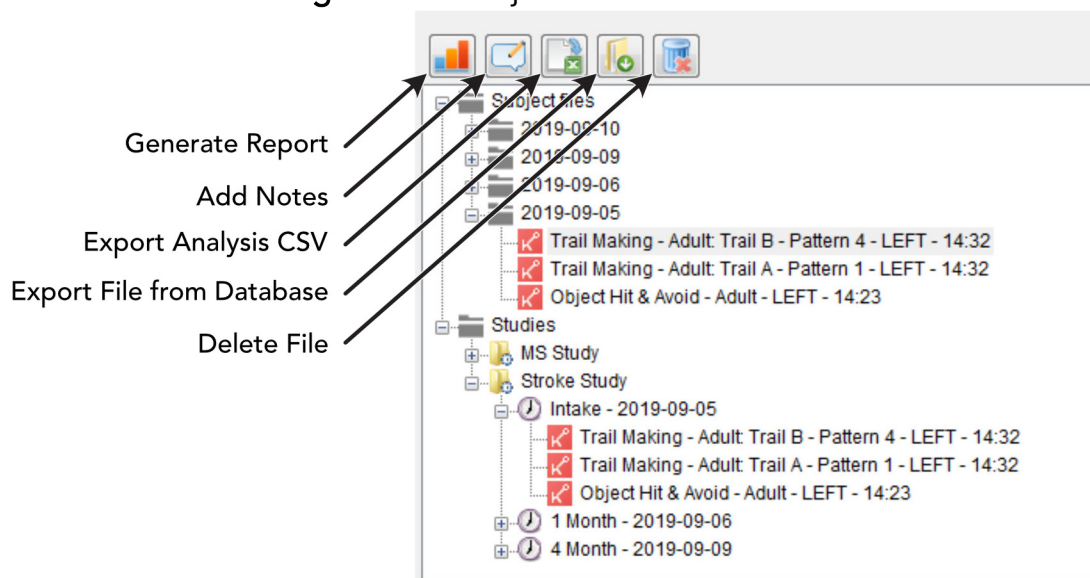
The other parts that are likely to require cleaning between uses include:

- The semi-transparent mirror
- The subject display
- Other horizontal surfaces

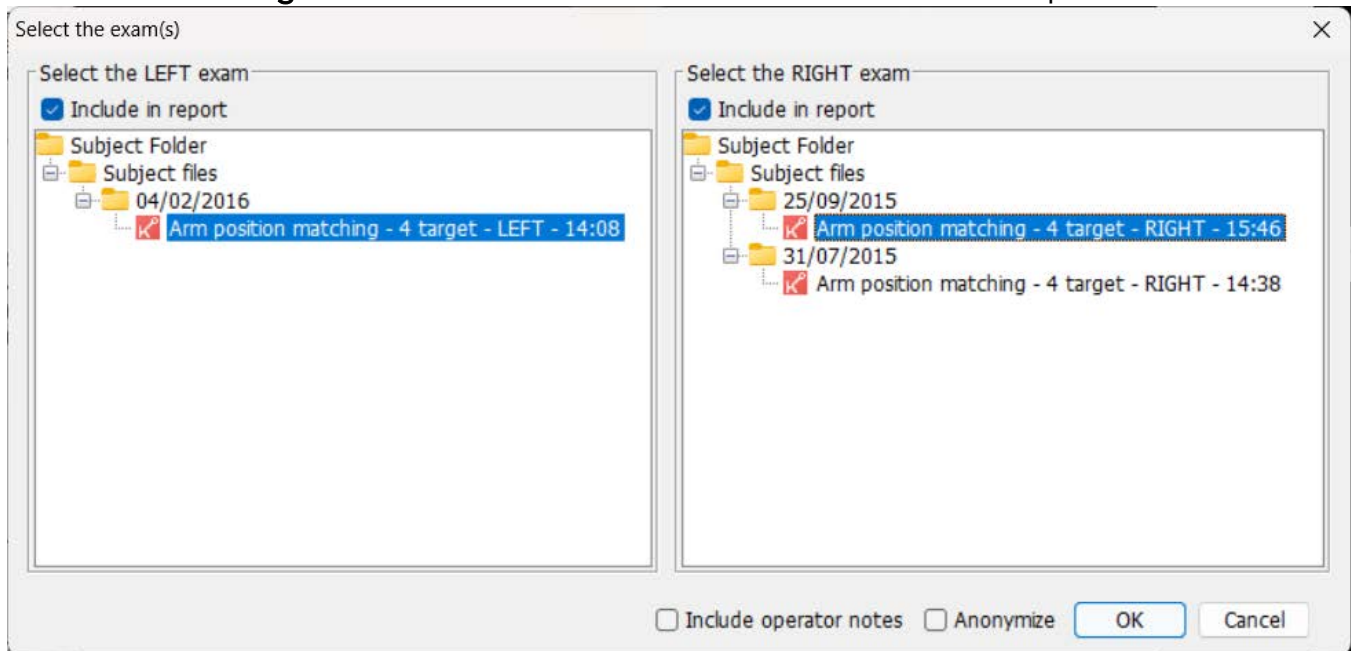
9.3 Generate a Standard Report

After a subject has performed a Standard Task, the data are available for reporting. In addition to Task Reports, Session and Longitudinal Reports can also be generated for these data using Dexterit-E Explorer. Session and Longitudinal Reports provide information from multiple KST in a single report, either for one session or across sessions. Please see the Dexterit-E Explorer User's Guide for more details. The following describes briefly how to generate a Task Report for a subject (for more details, refer to the Dexterit-E User Guide or to the Dexterit-E Explorer User Guide).

Figure 9.1: Subject Files Tree Controls



1. Select the desired subject in the subject list (see [Figure: 6.1](#)).
2. Select the data file from the Subject Files Tree for which you would like to generate a report.
3. Click the **Generate Report** button (see [Figure: 9.1](#)).
The *Select Exams* dialog opens.
4. In the *Select Exams* dialog, select the desired data file(s) from the available list.
The data files are sorted by date (see [Figure: 9.2](#)).

Figure 9.2: Selection of Data Files for the Dexterit-E Report

5. Check **Include operator notes** if you would like to include them.
6. If you would like the report to not contain a subject's name and date of birth, check **Anonymize**.
7. Click **OK** to generate a report.
If the data files have not yet been analyzed then the analysis will happen automatically, prior to report generation. Once the report is generated it will be added to the Subject Files Tree and it will be opened for viewing. See the Dexterit-E User Guide for full description of the contents of the Task Reports for each Standard Test.

9.4 Export a Standard Test Analysis to CSV Format

Standard Test analyses can be exported for each Standard Task to a CSV file. Each task has a unique set of parameters that are generated by the analysis and saved in the CSV file. See the Dexterit-E User Guide or KST Summary Guide for the list of common and task specific column names in the CSV file.

Figure 9.3: Export a Standard Test Analysis in Dexterit-E

Export...

Select KST data files to analyze.

1. Choose subjects for export

☐ Use all subjects

☒ Build a subject filter

☐ ID search

☐ Age range

☐ Study

☐ P.I.

☐ Pathology

☐ Exam date range

☐ Robot Type

☐ Provide subject ID's from a CSV file

☐ Manually choose subjects

2. Select exam type for export

☐ Arm Movement Matching

☐ Arm Position Matching

☐ Arm Posture Perturbation

☐ Ball on Bar

☐ Elbow Stretch

☐ Object Hit

☐ Object Hit & Avoid

☒ Paired Associates Learning

☐ Reverse Visually Guided Reaching

☐ Spatial Span

Trail Making

☐ Trail Making A+B

☐ Trail Making A

☐ Trail Making B

☐ Verification

☐ Visually Guided Reaching

Interlimb exports

☐ Interlimb - Elbow Stretch

☐ Interlimb - Reverse Visually Guided Reaching

☐ Interlimb - Visually Guided Reaching

☐ Clear existing analysis

☐ Use athlete normals

Destination

C:\Users\dmclean\Desktop\tmp\Paired Associates Learning.csv

☐ Anonymize - Removes the subject ID, name and date of birth

Export Cancel

1. Working in either Dexterit-E or Dexterit-E Explorer, select **File > Export > KST analysis as CSV**.
2. On the left of the dialog ([Figure: 9.3](#)), you can apply filters to automatically create a subset of subject data that meets the desired criteria or manually select subjects from the subject list.
 - If using a filter, check **Build a subject filter** and then check the attribute(s) that you wish to filter on. Enter or select an attribute value from the list.

- If selecting subjects, check **Manually choose subjects** and then check the appropriate subject(s).
- To remove personal information from the CSV file, check **Anonymize** and the subjects' names and dates of birth will be removed from the output CSV file.

The analysis results for the specified data will be exported to a CSV file for use in any spreadsheet software.

3. On the right of the dialog ([Figure: 9.3](#)), select the task that you wish to export the analysis results for.

Only one task may be selected at a time.

4. Click **Export** to start the export process

A CSV file will be generated, containing the KST Analysis results for the task and data files specified.

9.5 Data Replay

All data collected by Kinarm Labs, including that collected by a Kinarm Standard Task, are available for review by the user using Dexterit-E Explorer™. See the Dexterit-E Explorer User Guide for further details.

9.6 Data Analysis

All data collected by Kinarm Labs, including that collected by a Kinarm Standard Task, is available for further analysis by the user. Most users use MATLAB for performing analysis. See the Dexterit-E User Guide for further details.

9.7 Manage the Subject Database

The subject database includes all data collected as part of a Kinarm Lab. The data includes:

- Subject records (e.g. name, date of birth, etc)
- Data files (i.e. one data file for each task that was run)
- Sessions (i.e. groupings of data collected as part of a Study Protocol)
- Standard reports (optionally generated for KST)
- KST Analysis (auto-generated when a report is generated for KST)

Depending on the type of data, it can be viewed, edited, deleted, analyzed or exported. Some examples of editing features include:

- Re-naming data files or sessions
- Moving data files to a different subject or different session (i.e. if data were collected incorrectly)
- Adding notes to a data file

Some features are available in Dexterit-E, other features are available in Dexterit-E Explorer. See the applicable user guide for more details.

ATTENTION: Deleting data from the database cannot be undone.

9.8 Shut Down the Lab

The steps involved in shutting down the Kinarm Lab are very similar to those involved in powering it on (see [5.1 Power On the Kinarm Lab](#)). While it is recommended to power off various components to ensure longevity of the equipment, the Dexterit-E Computer may need to be left on to allow for data access from outside of the lab. All software should be properly shutdown, before powering off the equipment. The order in which to power off different pieces of equipment does not matter.

The different computers turn off using different methods:

- Dexterit-E Computer: using standard Windows power-down procedure
- Robot Computer: either using the button on the front of the computer or for newer systems with Power Console 15050 (as indicated on front label of Power Console), the switch on the Power Console will power off the Robot Computer.
- Gaze Tracking computer: using the power- off button within the computer operating system.

ATTENTION: If using the Deluxe Chair, charge the battery daily to maintain longevity of the battery. The battery should never be completely discharged.

1. Close all open software applications.
There should be no open software running on the Dexterit-E computer.
If applicable, the EyeLink software on the Gaze-Tracker Computer should be stopped.
2. Power off all equipment:
 - Kinarm Labs with Power Console 15050:
 - Power Console 15050
 - Dexterit-E Computer
 - Gaze-Tracker Computer

NOTE: The operator display will auto-shutdown a few minutes after the Dexterit-E Computer is powered down.

- Older Kinarm Labs:
 - Power Console and/or Robot(s)
 - Dexterit-E Computer
 - Robot Computer
 - Gaze-Tracker Computer

- Gaze power box

NOTE: The subject and operator displays will auto-shutdown a few minutes after the Dexter-E Computer is powered down.

3. If applicable, replace the lens cap on the Gaze-Tracker camera at the end of each session to keep dust off of the lens
4. If applicable, connect the charger to the charging port on the Deluxe Chair.
An orange LED on the charger indicates that charging is taking place.
The battery is charged when the LED on the charger changes to green.

10 Kinarm Lab Maintenance

There are a number of steps operators can take to prolong the life of the Kinarm Lab and maintain a clean environment for subjects. These actions are required each week the lab is in operation, and annually to look for wear and damage requiring repairs.

10.1 Weekly Checks

- Check for any visual wear and tear on the harness and belt straps and clips. Confirm that the clips and clasps work properly.
- Check that the gaze-tracking camera lens is clean. Wipe clean with a soft cloth.

ATTENTION: Take care to avoid disturbing the camera and/or illuminator while cleaning. If either is knocked out of position, the Kinarm Gaze-Tracker will not function correctly.

- Check that all extrusion end caps are present and in place.
- Check that unsupported software/OS is not installed or running on the Dexterit-E computer.
- The Dexterit-E Computer hard drive should be checked to ensure that there is sufficient space available to collect data. Available hard drive space is displayed in the status bar of Dexterit-E.

10.2 Annual Checks

- Robot
 - All Kinarm robot bearings should be checked by rotating each bearing with the Kinarm End-Point Lab turned off. Bearings should be smooth and silent. If they are not, the robot must be repaired.
 - Test the functionality of the Emergency Stop by depressing the button while active braking is enabled (i.e. a task is running, but the handles are not grasped). The robot should move freely, and the Emergency Stop and both Motor status LEDs on the back of the robots should turn red.
 - All visible screws should be checked and tightened on the Kinarm robot.
- Chair
 - The wheel brakes should be inspected to ensure that they are fully functional and have no signs of damage or wear.
 - For Deluxe Chair, check the seat lift for dirt, dust and stability:
 - Move the chair up to the highest position.
 - Clean the cylinder with a dry cloth. Do not use water or solvents.

- If the seat lift cylinder requires lubrication, apply a thin film of Teflon-based or silicon-based grease.
- For the Deluxe Chair, the battery must be replaced occasionally. The lifetime of the battery is typically 4-5 years, but depends highly on its level of use and whether it is charged daily as recommended by Kinarm. If the battery is several years old, or if it is showing signs of low capacity (i.e. only a few adjustments per charge), contact Kinarm Support for advice on replacing the battery.
- Virtual Reality Stand
 - Run the subject display calibration procedure to ensure correct video presentation to the subject.
- Computers and Software
 - Clean the dust from the computer fan filters to extend the lifetime of the computers. Slide the side panel of each computer backwards about 1.5" (it is not necessary to remove the entire panel) and pull out the removable filter that is resting behind the black mesh on the front of the computer. Gently wipe off any dust and then blow some air through the filter to remove any remaining dust. Reinstall the filter and the side panel.
 - Check the Kinarm Support web site for software updates.
 - It is recommended that an external data back-up policy be implemented to avoid data loss (i.e. in addition to the backup drive that is internal to the Dexterit-E Computer).
 - Scan the Dexterit-E hard drives for errors. In Windows Explorer, right-click on C: and select **Properties > Tools > Error checking**. If errors are found on the hard drive, allow the process to fix them automatically. Hard drive errors can be indicative of approaching system failure. Kinarm Support should be contacted immediately if errors are found.
 - Check both the main hard drive and the backup drive
 - Perform a manual back up of the Dexterit-E data using Dexterit-E's built-in backup feature and confirm that there are no errors (from the **File** menu, choose **Backup Database**).
- General
 - Inspect any external cables for signs of wear.
- Gaze-Tracker
 - The brightness of the illuminator should be checked to determine if it needs to be replaced. If pupil thresholds have dropped to be 70-80 on average, contact Kinarm Support. Like any other light source, the illuminator has limited lifetime and will eventually need to be replaced. The illuminator may need to be replaced after 3-5 years of typical use.